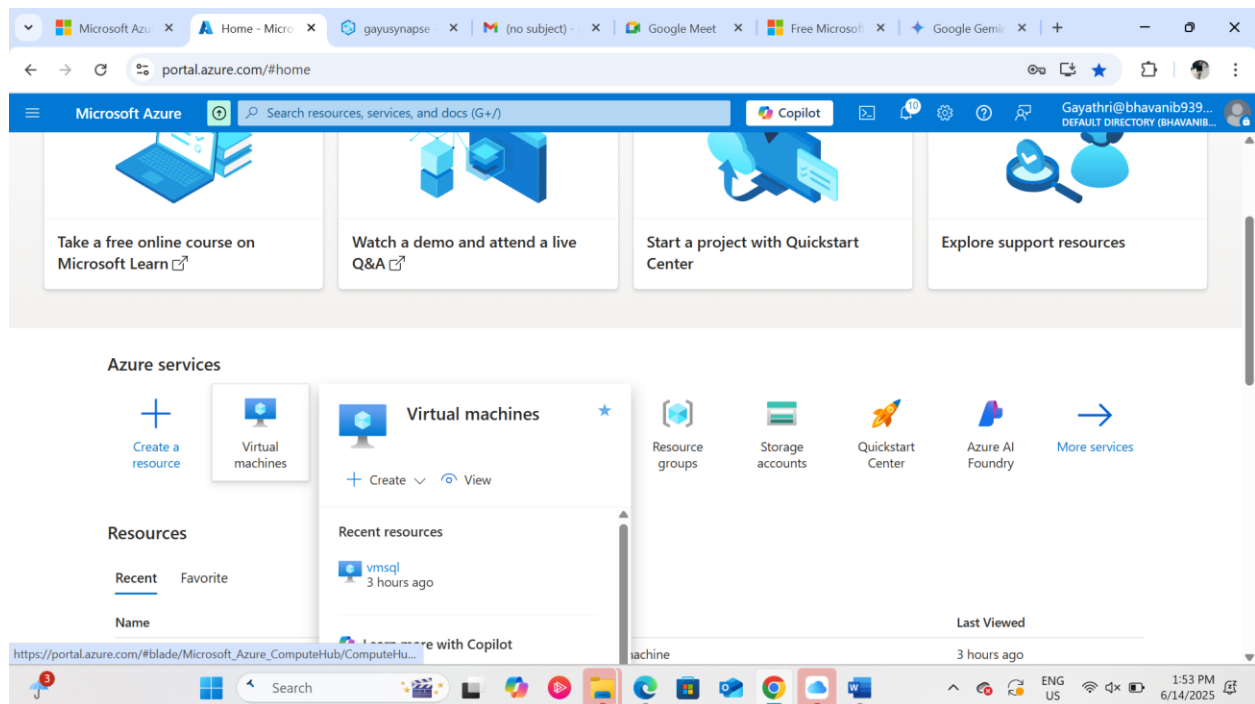


Design a dynamic pipeline that facilitates the following workflow:

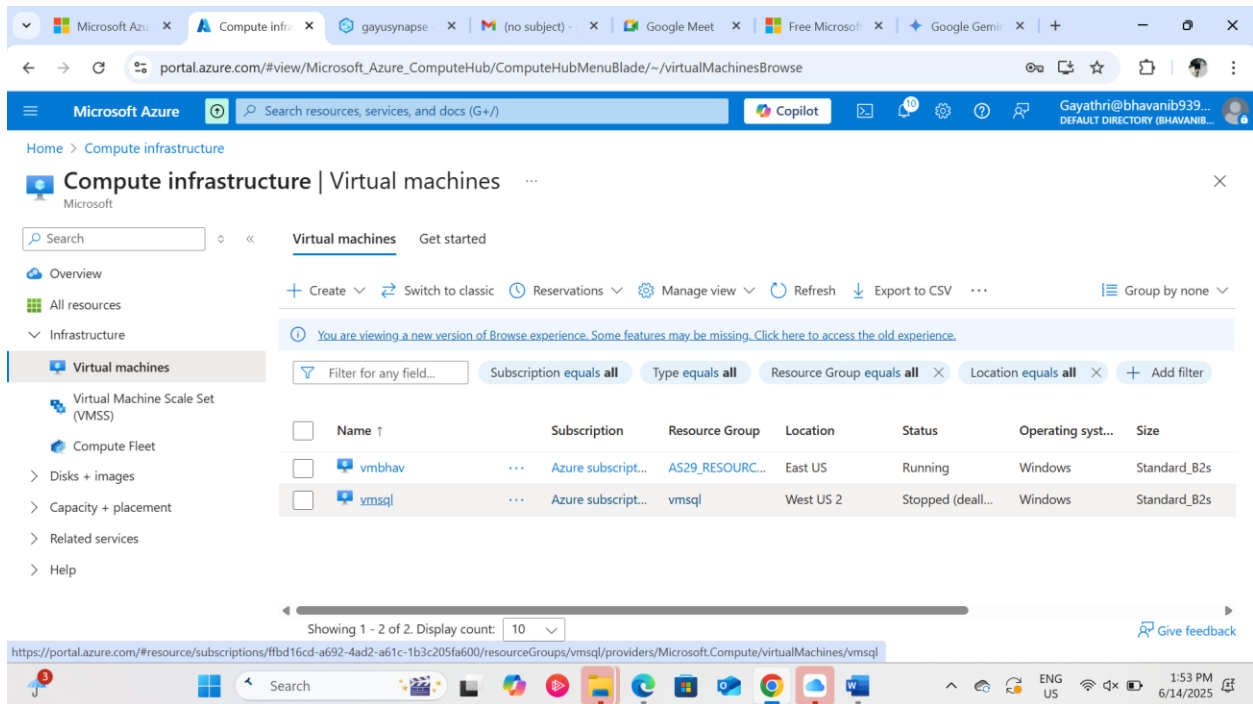
1. Extract data from 5 tables from an on-premises SQL Server.
2. Load the extracted data into Azure Data Lake Storage Gen2 in a structured format.
3. Copy the data from ADLS Gen2 back to a designated file system folder on the local C: drive.
4. Implement a metadata-driven approach using a control table that stores details such as:
 1. Source table names
 2. Target file paths
5. Implement the email configuration task by using logic apps and send an email for any failure.

This approach should enable scalable and automated data movement while maintaining centralized control through the metadata table.

Click on virtual Machine and click on create



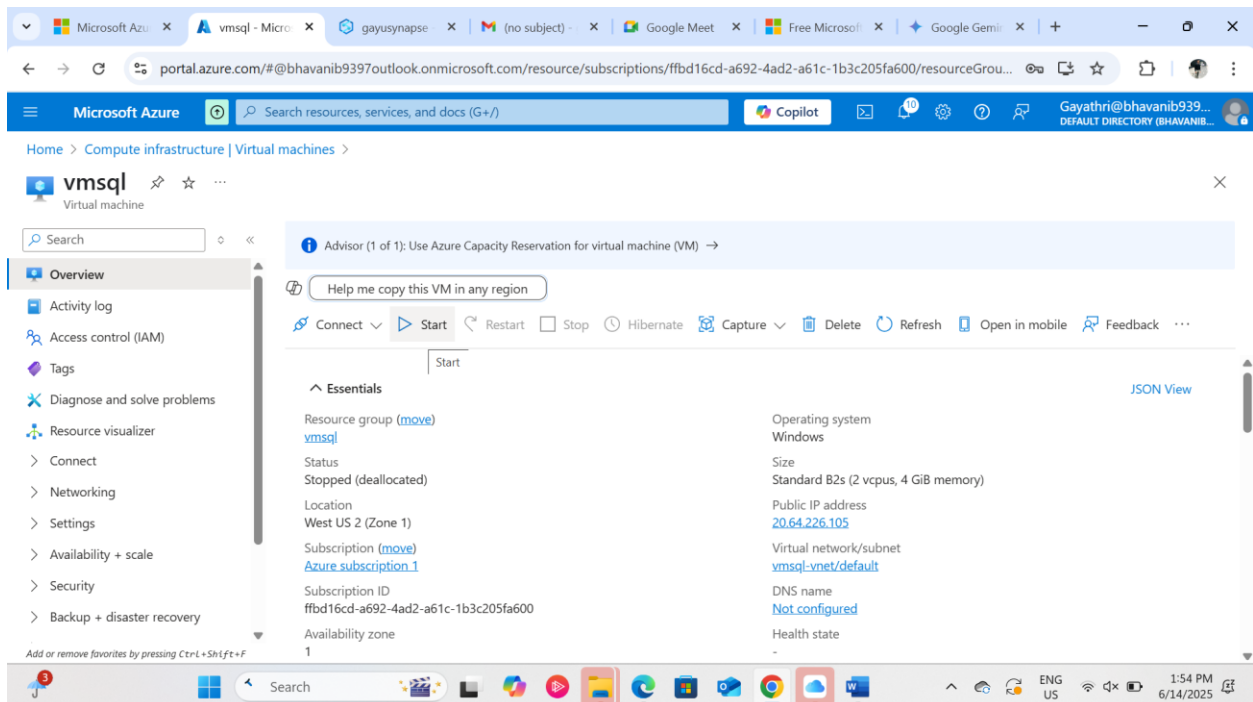
Click on Vmsql which is our virtual machine name



The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation links: Overview, All resources, Infrastructure, Virtual machines (selected), Virtual Machine Scale Set (VMSS), Compute Fleet, Disks + images, Capacity + placement, Related services, and Help. The main area displays the 'Virtual machines' page with a table of resources. The table has columns: Name, Subscription, Resource Group, Location, Status, Operating system, and Size. Two VMs are listed: 'vmbhav' (Running, Windows, Standard_B2s) and 'vmsql' (Stopped (deallocated), Windows, Standard_B2s). Below the table, it says 'Showing 1 - 2 of 2. Display count: 10'. The URL in the address bar is 'https://portal.azure.com/#resource/subscriptions/ffbd16cd-a692-4ad2-a61c-1b3c205fa600/resourceGroups/vmsql/providers/Microsoft.Compute/virtualMachines/vmsql'.

Name	Subscription	Resource Group	Location	Status	Operating system	Size
vmbhav	Azure subscript...	AS29_RESOURCE...	East US	Running	Windows	Standard_B2s
vmsql	Azure subscript...	vmsql	West US 2	Stopped (deall...	Windows	Standard_B2s

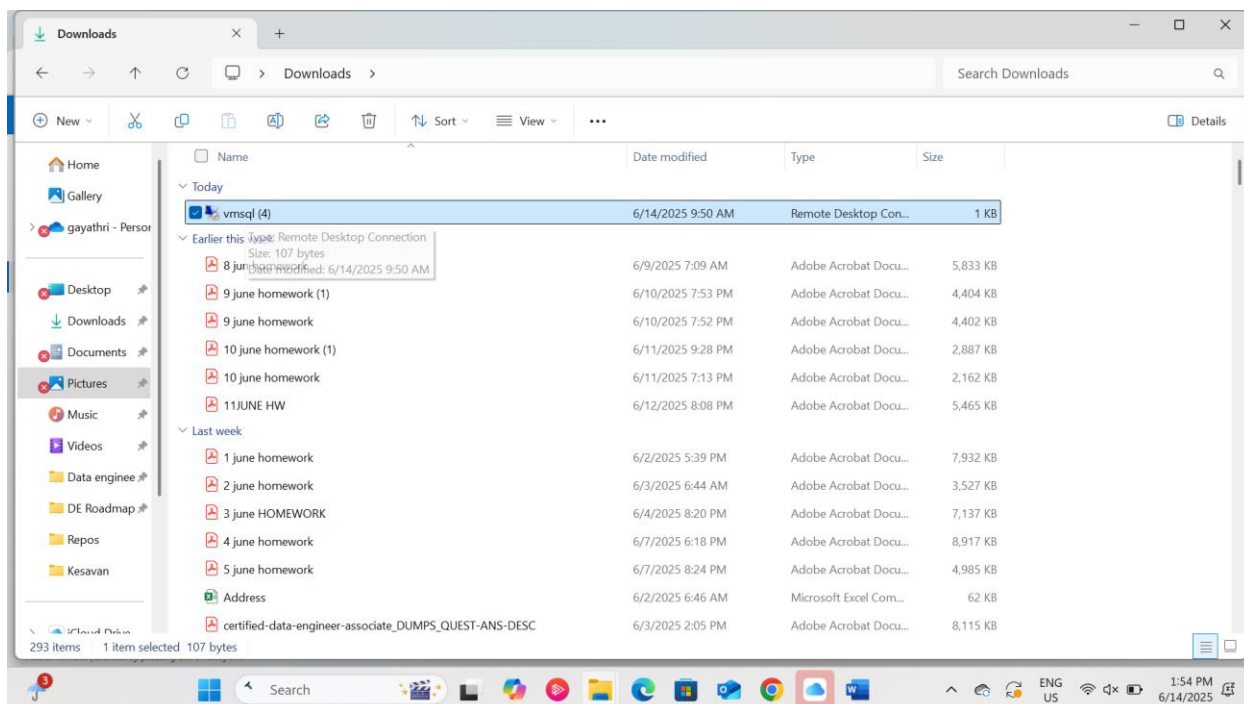
Start virtual machine, in this page



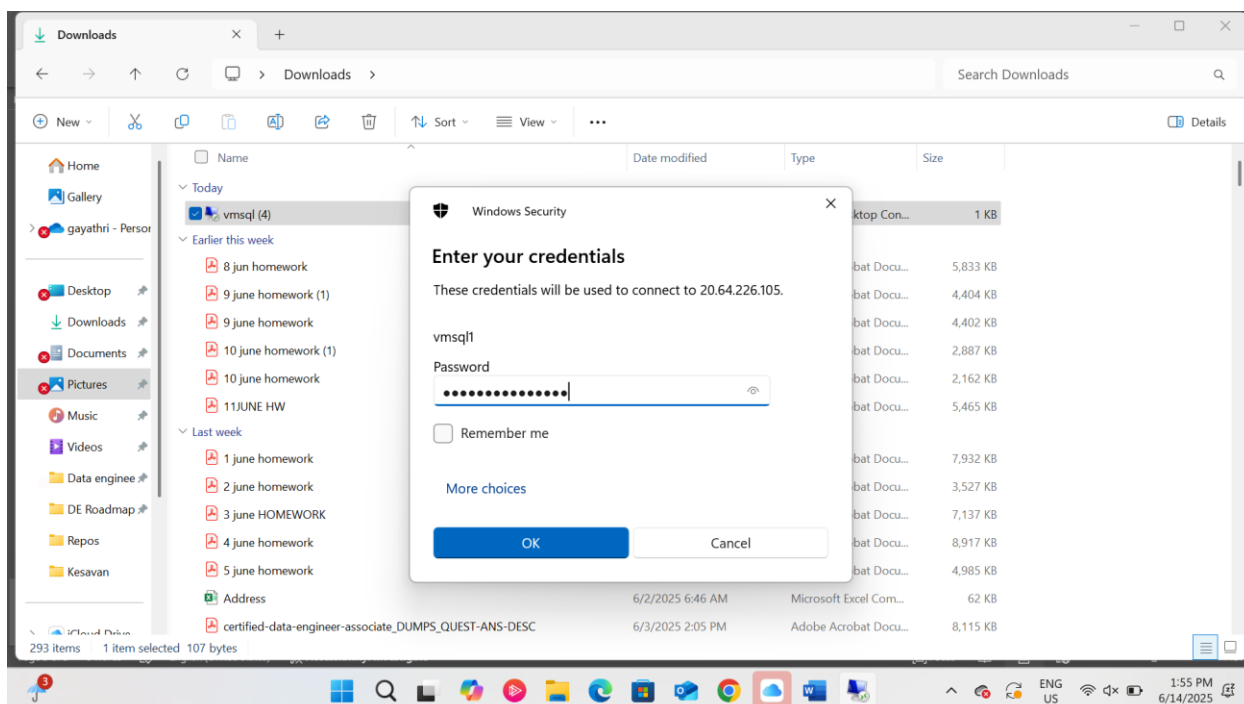
The screenshot shows the Microsoft Azure portal interface for the 'vmsql' virtual machine. The left sidebar contains navigation links: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Connect, Networking, Settings, Availability + scale, Security, and Backup + disaster recovery. The main area displays the 'vmsql' VM details page. At the top, there is an 'Advisor' tip: 'Use Azure Capacity Reservation for virtual machine (VM)'. Below this, there is a 'Help me copy this VM in any region' button. The 'Start' button is highlighted. The 'Essentials' section shows the following details: Resource group (vmsql), Status (Stopped (deallocated)), Location (West US 2 (Zone 1)), Subscription (Azure subscription 1), Subscription ID (ffbd16cd-a692-4ad2-a61c-1b3c205fa600), Availability zone (1), Operating system (Windows), Size (Standard B2s (2 vcpus, 4 GiB memory)), Public IP address (20.64.226.105), Virtual network/subnet (vmsql-vnet/default), DNS name (Not configured), and Health state (-). The URL in the address bar is 'https://portal.azure.com/#@bhavanib9397outlook.onmicrosoft.com/resource/subscriptions/ffbd16cd-a692-4ad2-a61c-1b3c205fa600/resourceGroups/vmsql/providers/Microsoft.Compute/virtualMachines/vmsql'.

Essentials	Details
Resource group (move)	vmsql
Status	Stopped (deallocated)
Location	West US 2 (Zone 1)
Subscription (move)	Azure subscription 1
Subscription ID	ffbd16cd-a692-4ad2-a61c-1b3c205fa600
Availability zone	1
Operating system	Windows
Size	Standard B2s (2 vcpus, 4 GiB memory)
Public IP address	20.64.226.105
Virtual network/subnet	vmsql-vnet/default
DNS name	Not configured
Health state	-

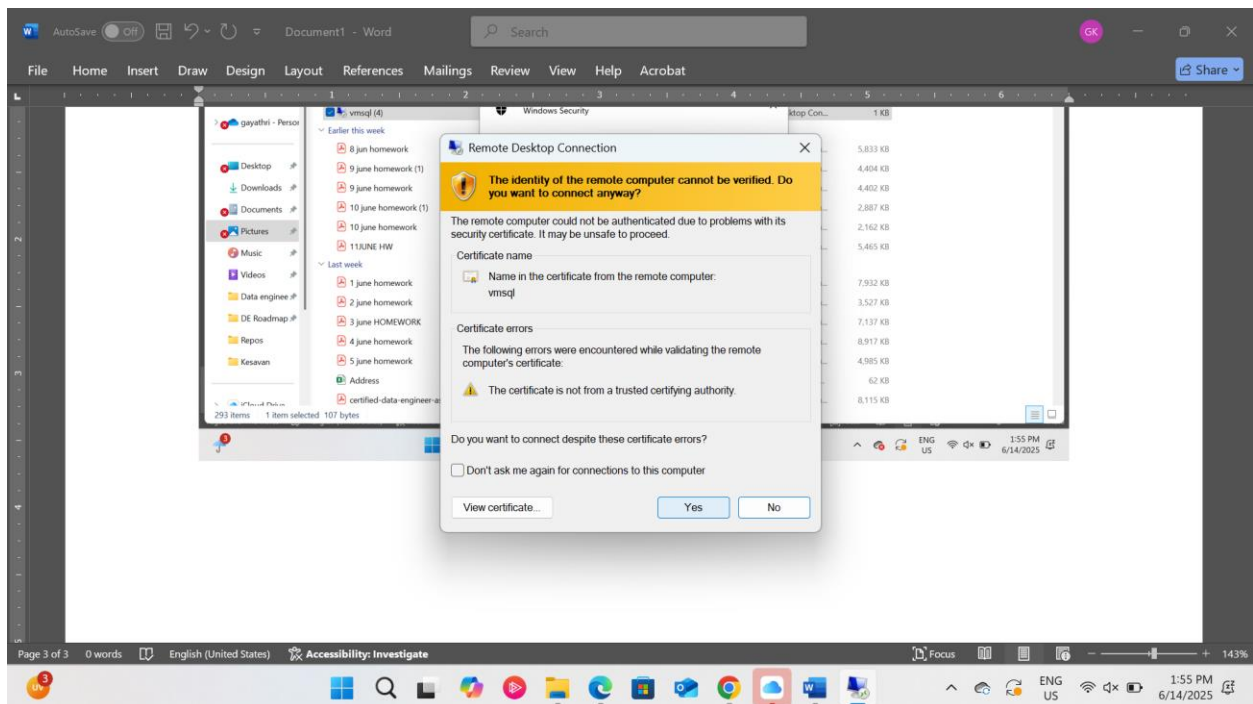
From the download, click on vmsql



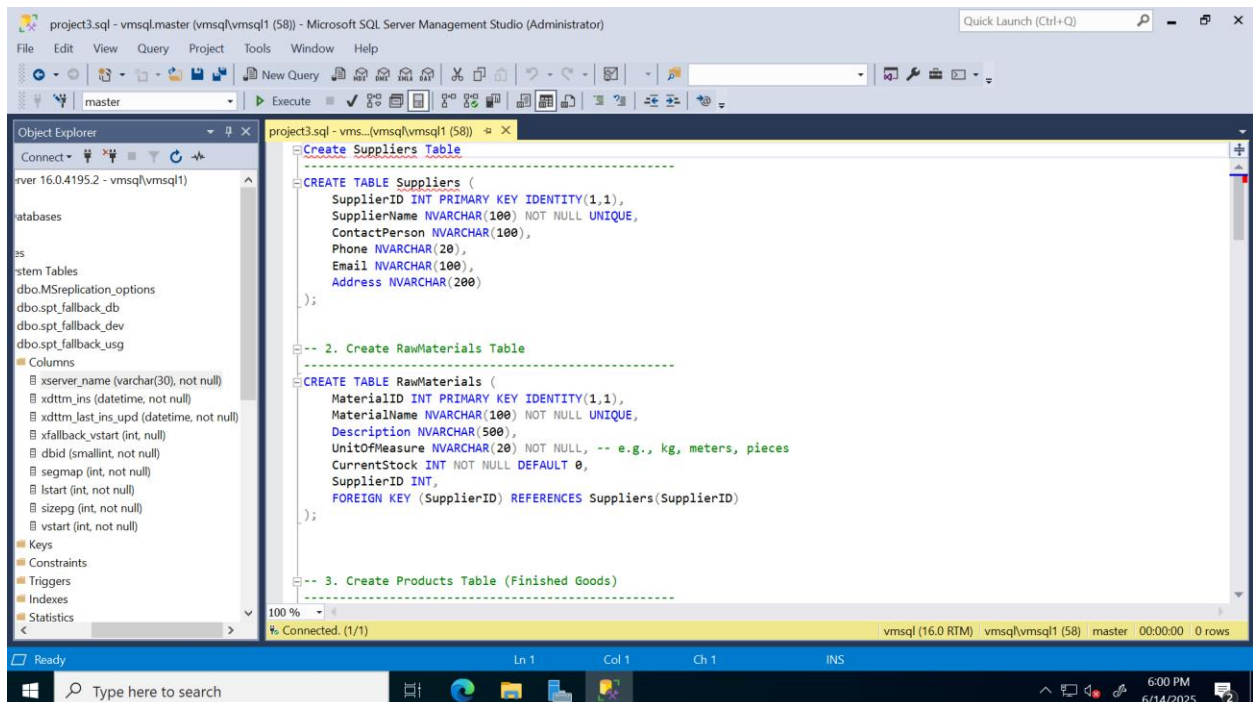
Type user name and password for vmsql1



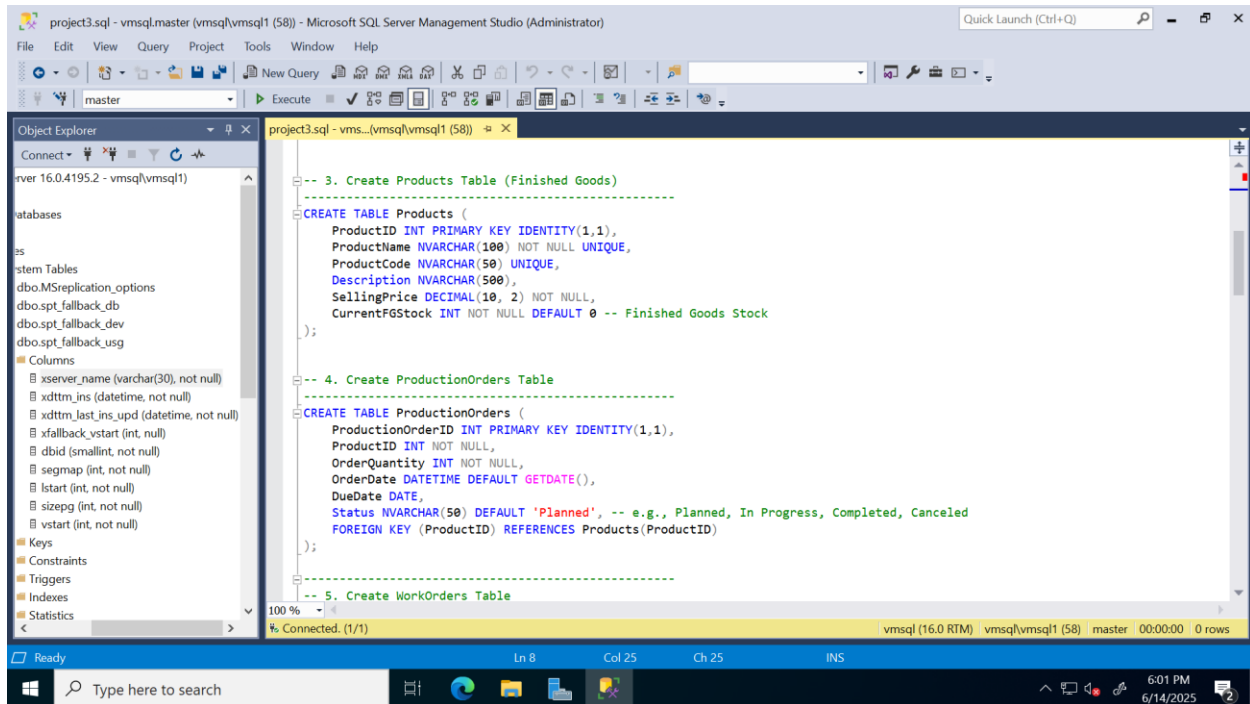
Click on yes, to open remote desktop



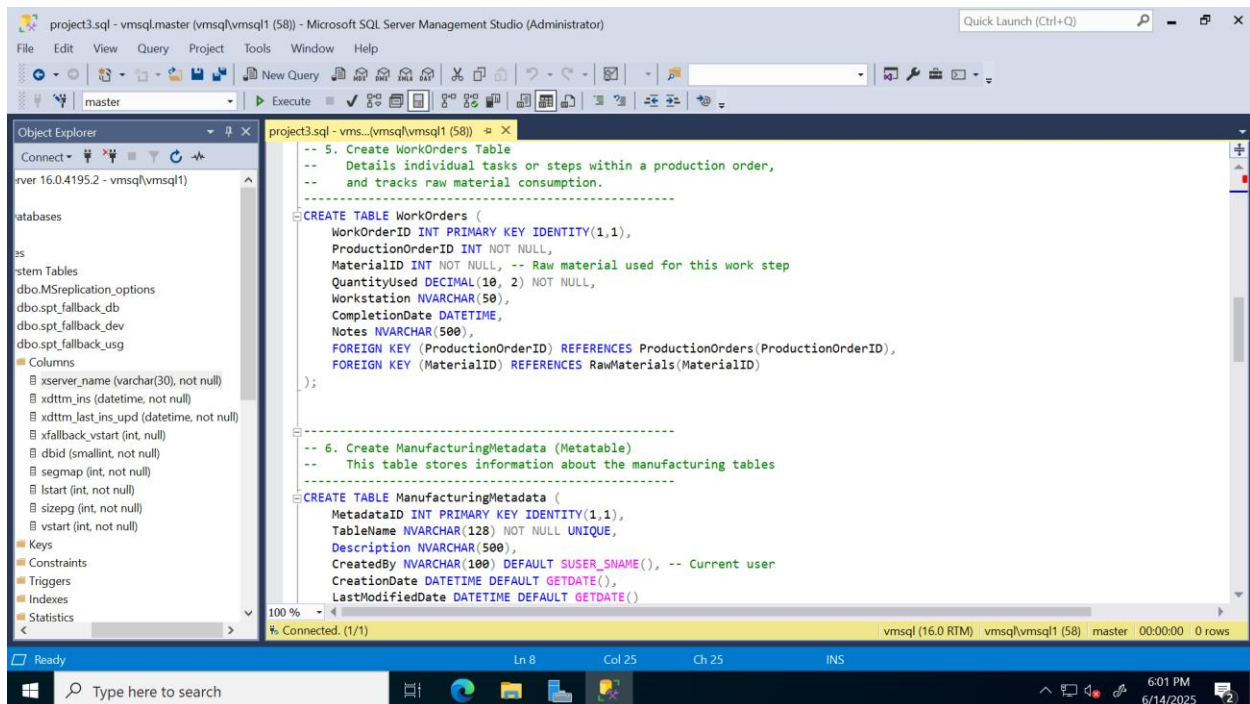
Create 5 table and insert values in the SSMS, Create table, Raw material table



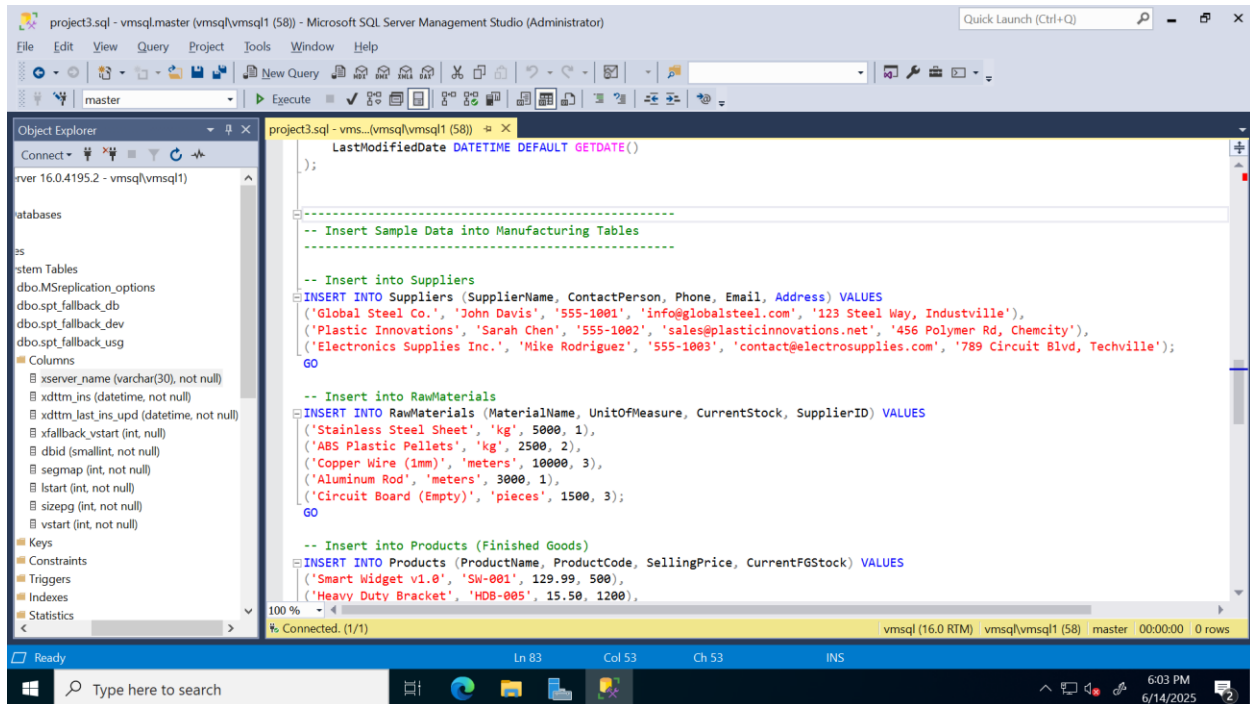
Create 5 table and insert values in the SSMS, Product table, production order table



Create 5 table and insert values in the SSMS, workorder table, ManufacturingMetadata



Insert values in table suppliers, Raw material, Products



The screenshot shows the Microsoft SQL Server Management Studio (SSMS) interface. The left pane displays the 'Object Explorer' with the 'Columns' view selected for the 'dbo.spt_fallback_dev' table. The main query window shows a SQL script with the following content:

```

-- Insert Sample Data into Manufacturing Tables

-- Insert into Suppliers
INSERT INTO Suppliers (SupplierName, ContactPerson, Phone, Email, Address) VALUES
('Global Steel Co.', 'John Davis', '555-1001', 'info@globalsteel.com', '123 Steel Way, Indusville'),
('Plastic Innovations', 'Sarah Chen', '555-1002', 'sales@plasticinnovations.net', '456 Polymer Rd, Chemcity'),
('Electronics Supplies Inc.', 'Mike Rodriguez', '555-1003', 'contact@electrosupplies.com', '789 Circuit Blvd, Techville');
GO

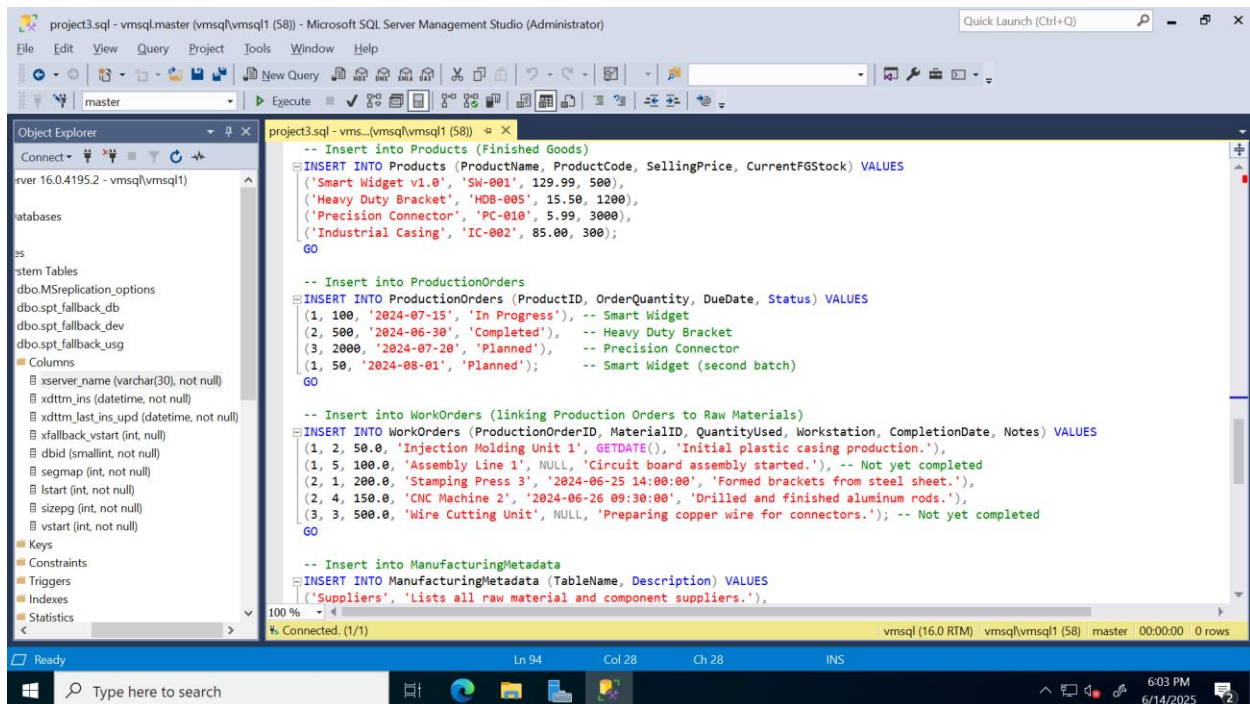
-- Insert into RawMaterials
INSERT INTO RawMaterials (MaterialName, UnitOfMeasure, CurrentStock, SupplierID) VALUES
('Stainless Steel Sheet', 'kg', 5000, 1),
('ABS Plastic Pellets', 'kg', 2500, 2),
('Copper Wire (1mm)', 'meters', 10000, 3),
('Aluminum Rod', 'meters', 3000, 1),
('Circuit Board (Empty)', 'pieces', 1500, 3);
GO

-- Insert into Products (Finished Goods)
INSERT INTO Products (ProductName, ProductCode, SellingPrice, CurrentFGStock) VALUES
('Smart Widget v1.0', 'SW-001', 129.99, 500),
('Heavy Duty Bracket', 'HDB-005', 15.50, 1200);
GO

```

The status bar at the bottom indicates the script is connected to 'vmsql (16.0 RTM)' and 'vmsql/vmsql1 (58)' master, with 0 rows affected.

Insert values in table production orders, work orders



The screenshot shows the Microsoft SQL Server Management Studio (SSMS) interface. The left pane displays the 'Object Explorer' with the 'Columns' view selected for the 'dbo.spt_fallback_dev' table. The main query window shows a SQL script with the following content:

```

-- Insert into Products (Finished Goods)
INSERT INTO Products (ProductName, ProductCode, SellingPrice, CurrentFGStock) VALUES
('Smart Widget v1.0', 'SW-001', 129.99, 500),
('Heavy Duty Bracket', 'HDB-005', 15.50, 1200),
('Precision Connector', 'PC-010', 5.99, 3000),
('Industrial Casing', 'IC-002', 85.00, 300);
GO

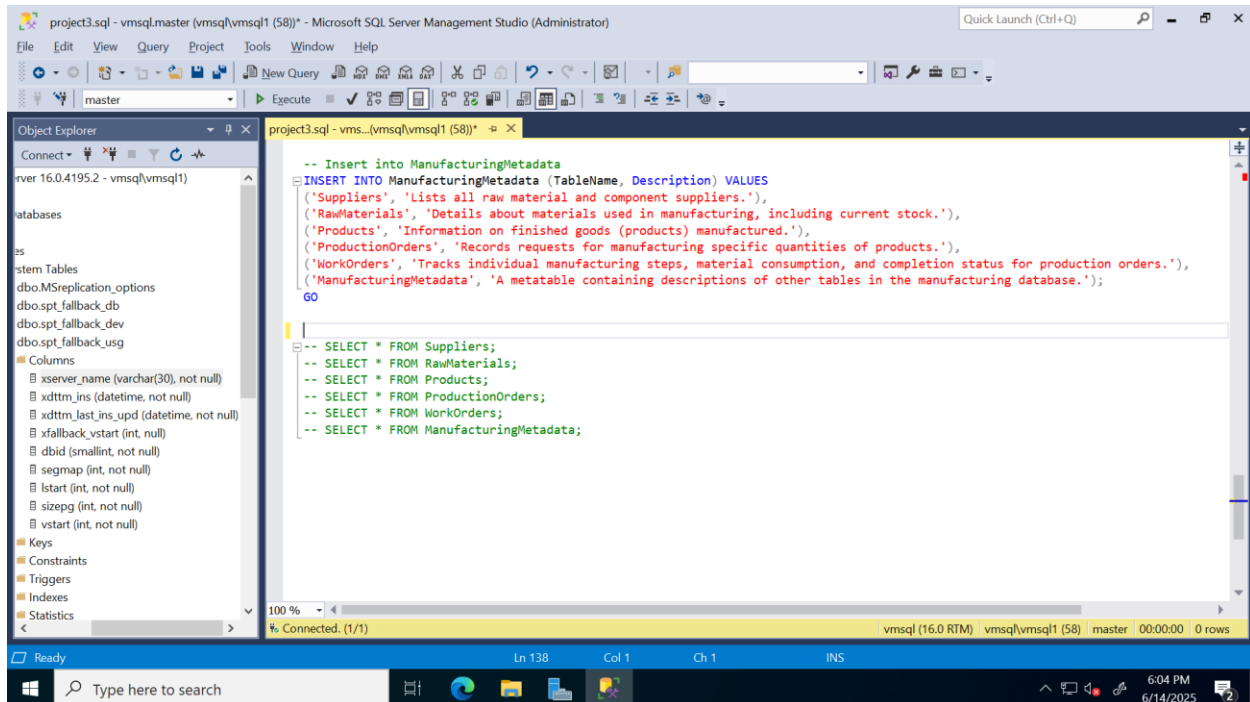
-- Insert into ProductionOrders
INSERT INTO ProductionOrders (ProductID, OrderQuantity, DueDate, Status) VALUES
(1, 100, '2024-07-15', 'In Progress'), -- Smart Widget
(2, 500, '2024-06-30', 'Completed'), -- Heavy Duty Bracket
(3, 2000, '2024-07-20', 'Planned'), -- Precision Connector
(1, 50, '2024-08-01', 'Planned'); -- Smart Widget (second batch)
GO

-- Insert into WorkOrders (Linking Production Orders to Raw Materials)
INSERT INTO WorkOrders (ProductionOrderID, MaterialID, QuantityUsed, Workstation, CompletionDate, Notes) VALUES
(1, 2, 50.0, 'Injection Molding Unit 1', GETDATE(), 'Initial plastic casing production.'),
(1, 5, 100.0, 'Assembly Line 1', NULL, 'Circuit board assembly started.'), -- Not yet completed
(2, 1, 200.0, 'Stamping Press 3', '2024-06-25 14:00:00', 'Formed brackets from steel sheet.'),
(2, 4, 150.0, 'CNC Machine 2', '2024-06-26 09:30:00', 'Drilled and finished aluminum rods.'),
(3, 3, 500.0, 'Wire Cutting Unit', NULL, 'Preparing copper wire for connectors.'); -- Not yet completed
GO

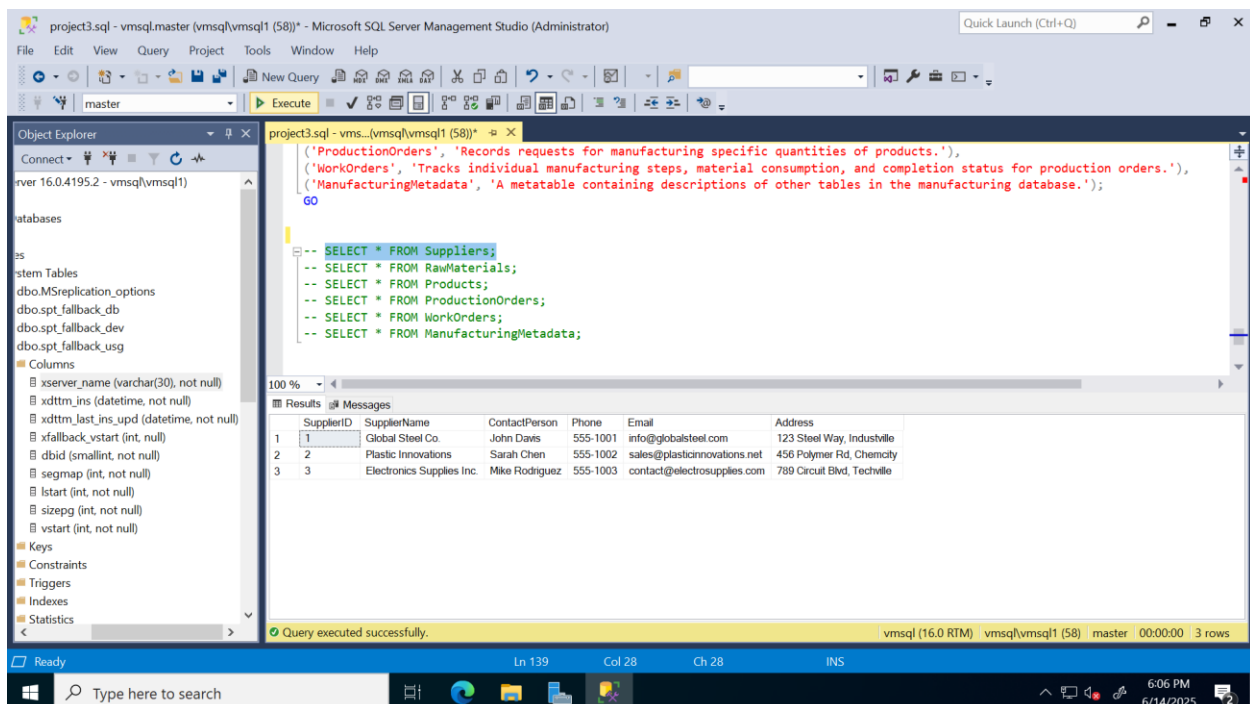
-- Insert into ManufacturingMetadata
INSERT INTO ManufacturingMetadata (TableName, Description) VALUES
('Suppliers', 'lists all raw material and component suppliers.');
```

The status bar at the bottom indicates the script is connected to 'vmsql (16.0 RTM)' and 'vmsql/vmsql1 (58)' master, with 0 rows affected.

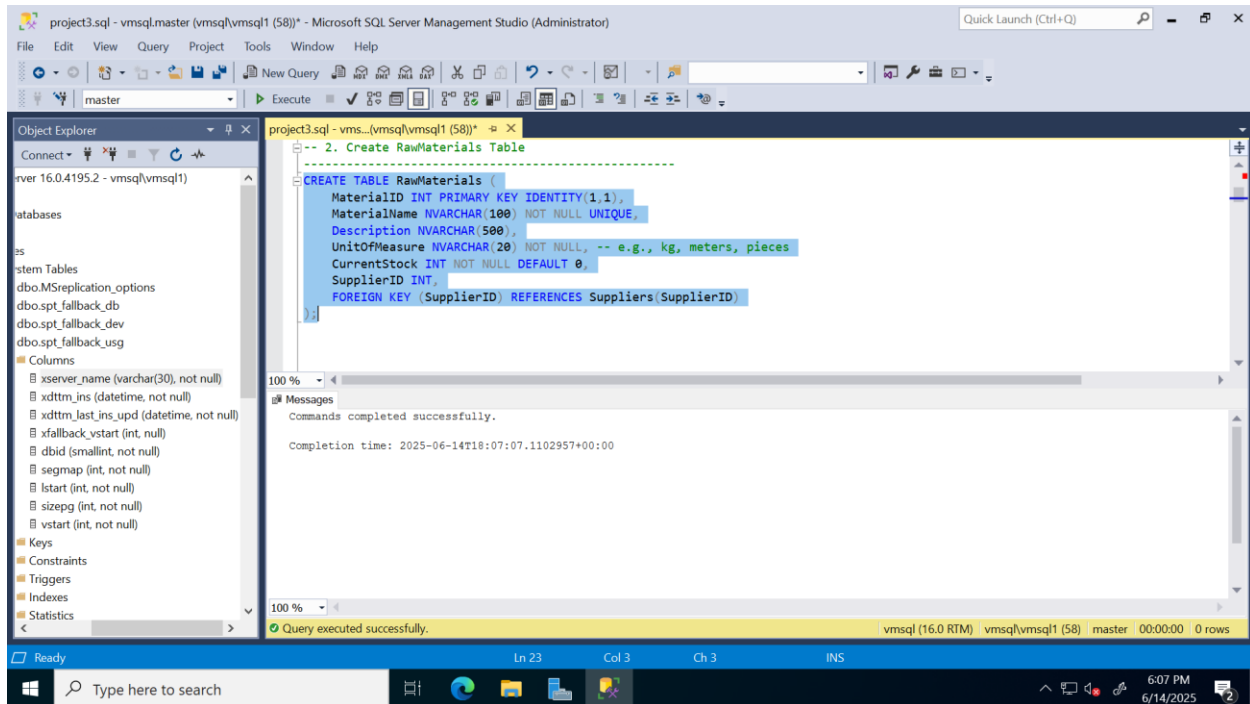
Insert values in table for ManufacturingMetadata



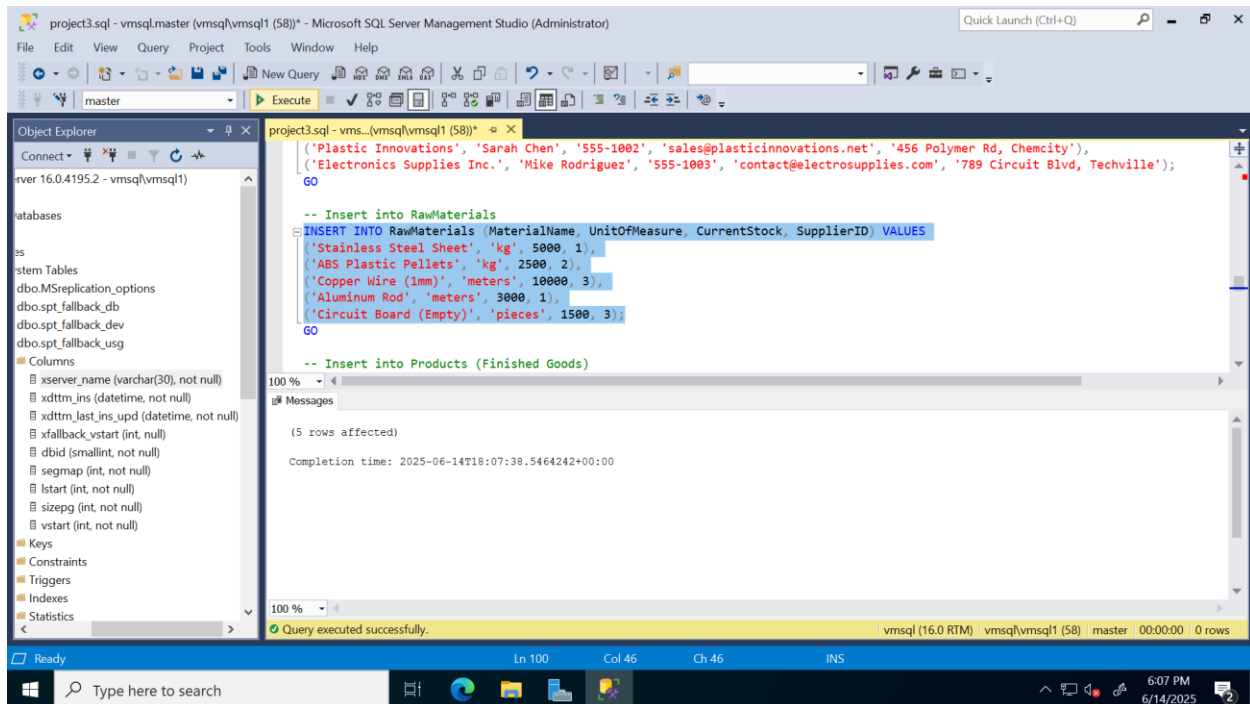
Select the values from suppliers, Raw materials, products, Production Orders, Work Orders, Manufacturing Metadata and executed the suppliers, it is successful.



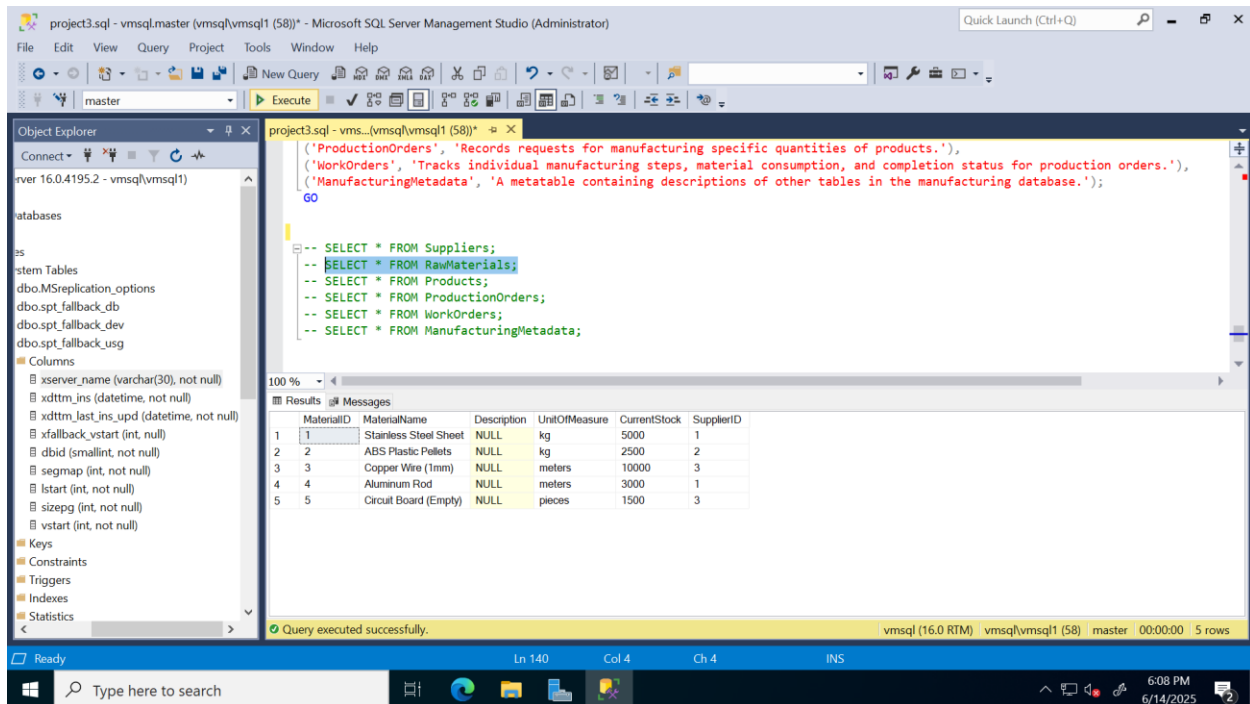
Executed the Raw material it is successful.



Inserted value in Raw materials and it is successful



Executed the Raw materials and the table is created successful



project3.sql - vmsql.master (vmsql\vmsql1 (58)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Tools Window Help

master

Execute

Object Explorer

Connect +

Server 16.0.4195.2 - vmsql\vmsql1

atabases

System Tables

dbo.MSReplication_options

dbo.spt_fallback_db

dbo.spt_fallback_dev

dbo.spt_fallback_usg

Columns

xserver_name (varchar(30), not null)

xdtm_ins (datetime, not null)

xdtm_last_ins_upd (datetime, not null)

xfallback_vstart (int, null)

dbid (smallint, not null)

segmap (int, not null)

lstart (int, not null)

sizepg (int, not null)

vstart (int, not null)

Keys

Constraints

Triggers

Indexes

Statistics

project3.sql - vms... (vmsql\vmsql1 (58))

```

-- ('ProductionOrders', 'Records requests for manufacturing specific quantities of products.').
-- ('WorkOrders', 'Tracks individual manufacturing steps, material consumption, and completion status for production orders.').
-- ('ManufacturingMetadata', 'A metatable containing descriptions of other tables in the manufacturing database.').
GO

-- SELECT * FROM Suppliers;
-- SELECT * FROM RawMaterials;
-- SELECT * FROM Products;
-- SELECT * FROM ProductionOrders;
-- SELECT * FROM WorkOrders;
-- SELECT * FROM ManufacturingMetadata;

```

100 %

Results

MaterialID	MaterialName	Description	UnitOfMeasure	CurrentStock	SupplierID
1	Stainless Steel Sheet	NULL	kg	5000	1
2	ABS Plastic Pellets	NULL	kg	2500	2
3	Copper Wire (1mm)	NULL	meters	10000	3
4	Aluminum Rod	NULL	meters	3000	1
5	Circuit Board (Empty)	NULL	pieces	1500	3

Messages

Query executed successfully.

vmsql (16.0 RTM) vmsql\vmsql1 (58) master 00:00:00 5 rows

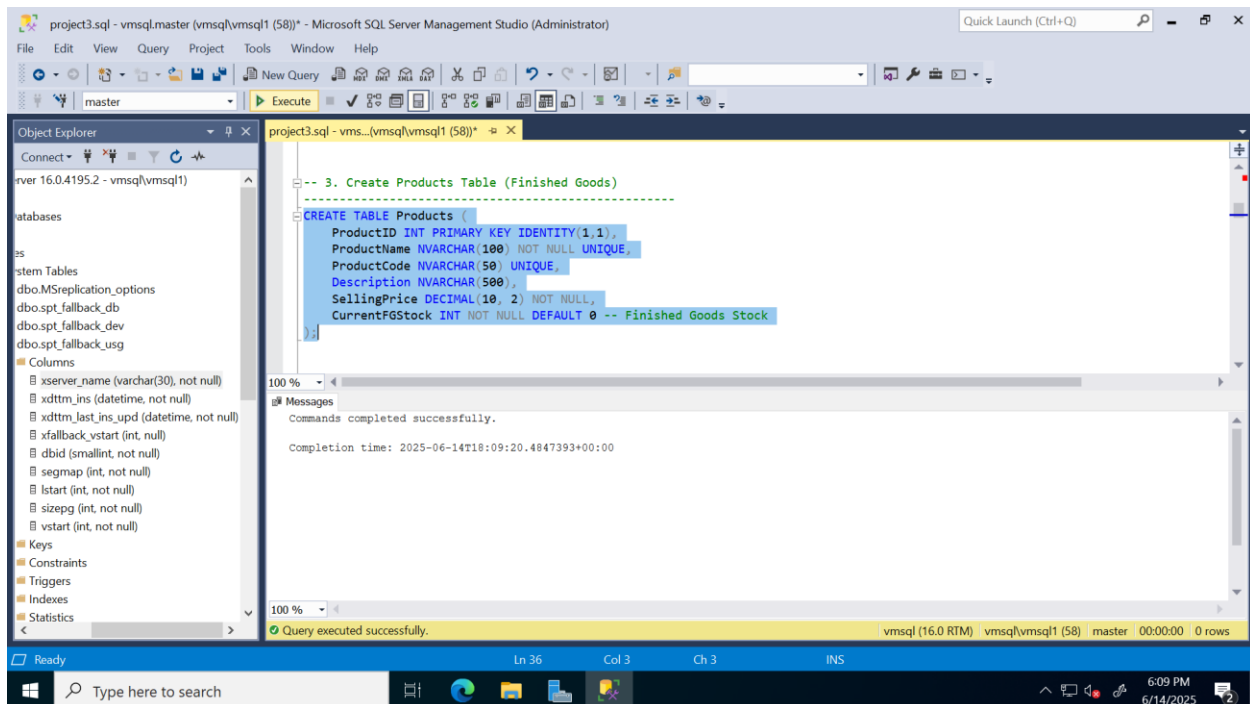
Ready

Type here to search

Ln 140 Col 4 Ch 4 INS

6:08 PM 6/14/2025

Create table products and it is executed successfully



project3.sql - vmsql.master (vmsql\vmsql1 (58)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Tools Window Help

master

Execute

Object Explorer

Connect +

Server 16.0.4195.2 - vmsql\vmsql1

atabases

System Tables

dbo.MSReplication_options

dbo.spt_fallback_db

dbo.spt_fallback_dev

dbo.spt_fallback_usg

Columns

xserver_name (varchar(30), not null)

xdtm_ins (datetime, not null)

xdtm_last_ins_upd (datetime, not null)

xfallback_vstart (int, null)

dbid (smallint, not null)

segmap (int, not null)

lstart (int, not null)

sizepg (int, not null)

vstart (int, not null)

Keys

Constraints

Triggers

Indexes

Statistics

project3.sql - vms... (vmsql\vmsql1 (58))

```

-- 3. Create Products Table (Finished Goods)
-- -----
CREATE TABLE Products (
    ProductID INT PRIMARY KEY IDENTITY(1,1),
    ProductName NVARCHAR(100) NOT NULL UNIQUE,
    ProductCode NVARCHAR(50) UNIQUE,
    Description NVARCHAR(500),
    SellingPrice DECIMAL(10, 2) NOT NULL,
    CurrentFGStock INT NOT NULL DEFAULT 0 -- Finished Goods Stock
);

```

100 %

Messages

Commands completed successfully.

Completion time: 2025-06-14T18:09:20.4847393+00:00

Query executed successfully.

vmsql (16.0 RTM) vmsql\vmsql1 (58) master 00:00:00 0 rows

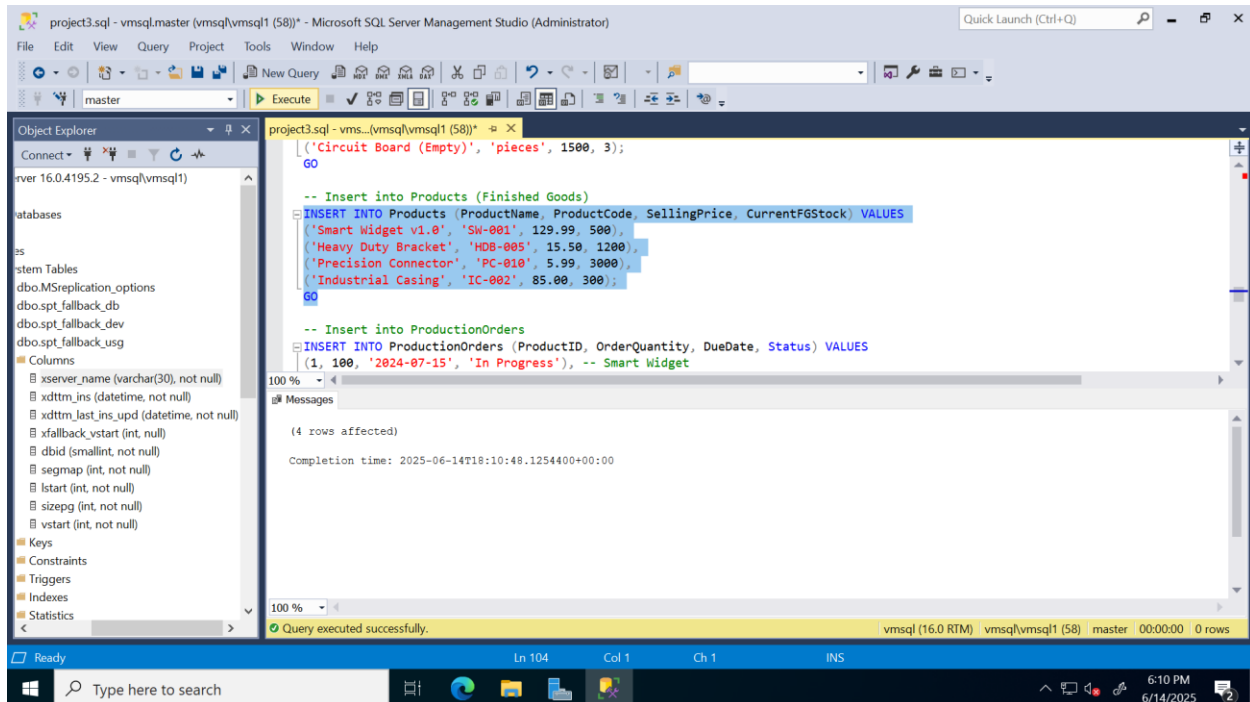
Ready

Type here to search

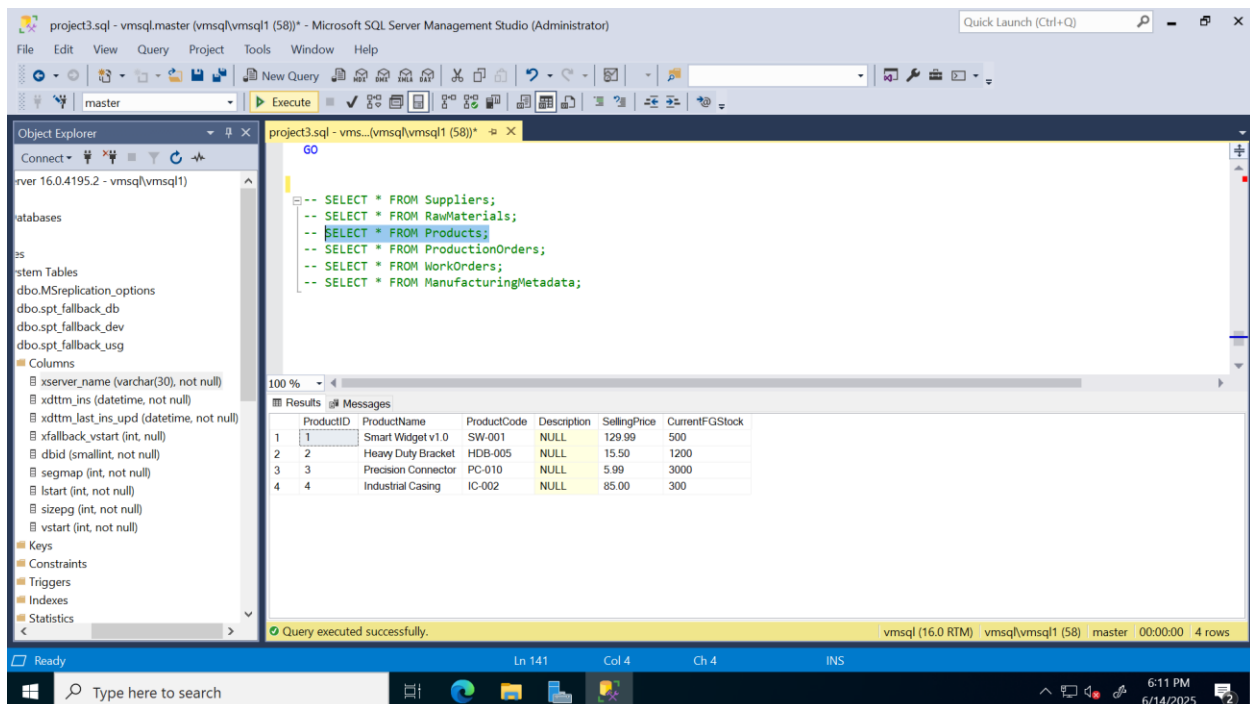
Ln 36 Col 3 Ch 3 INS

6:09 PM 6/14/2025

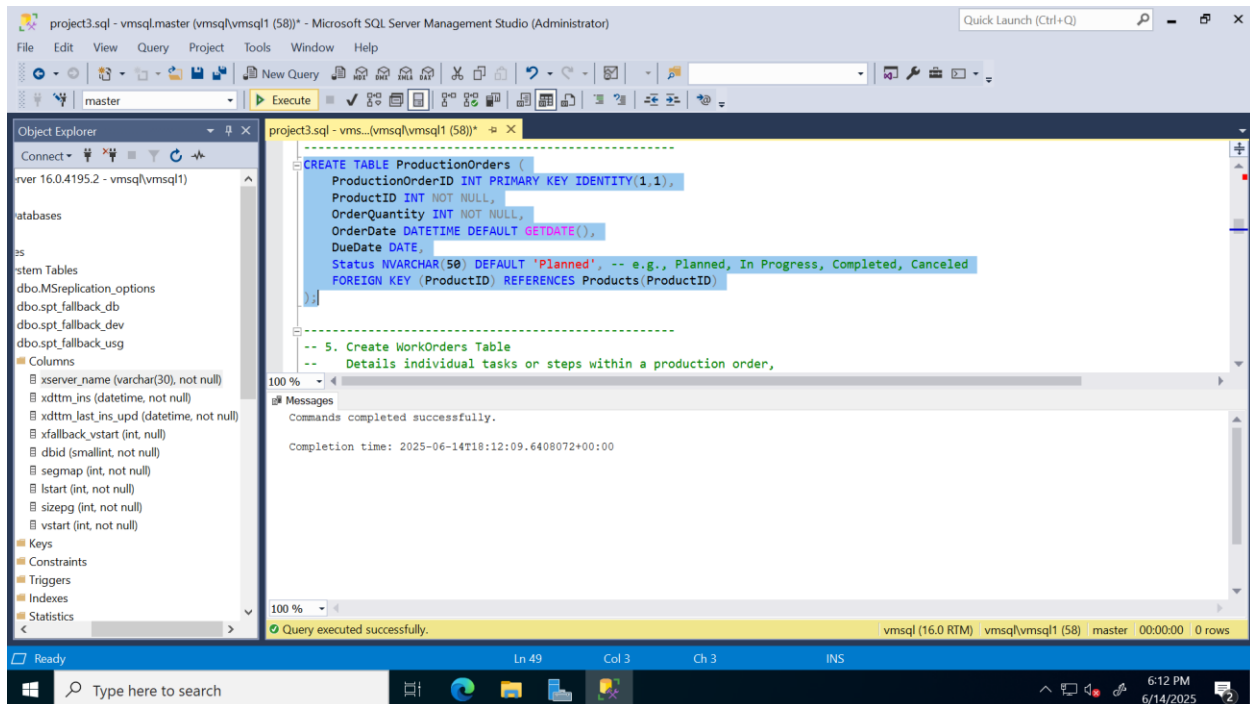
Inserted value in products and executed it is successful



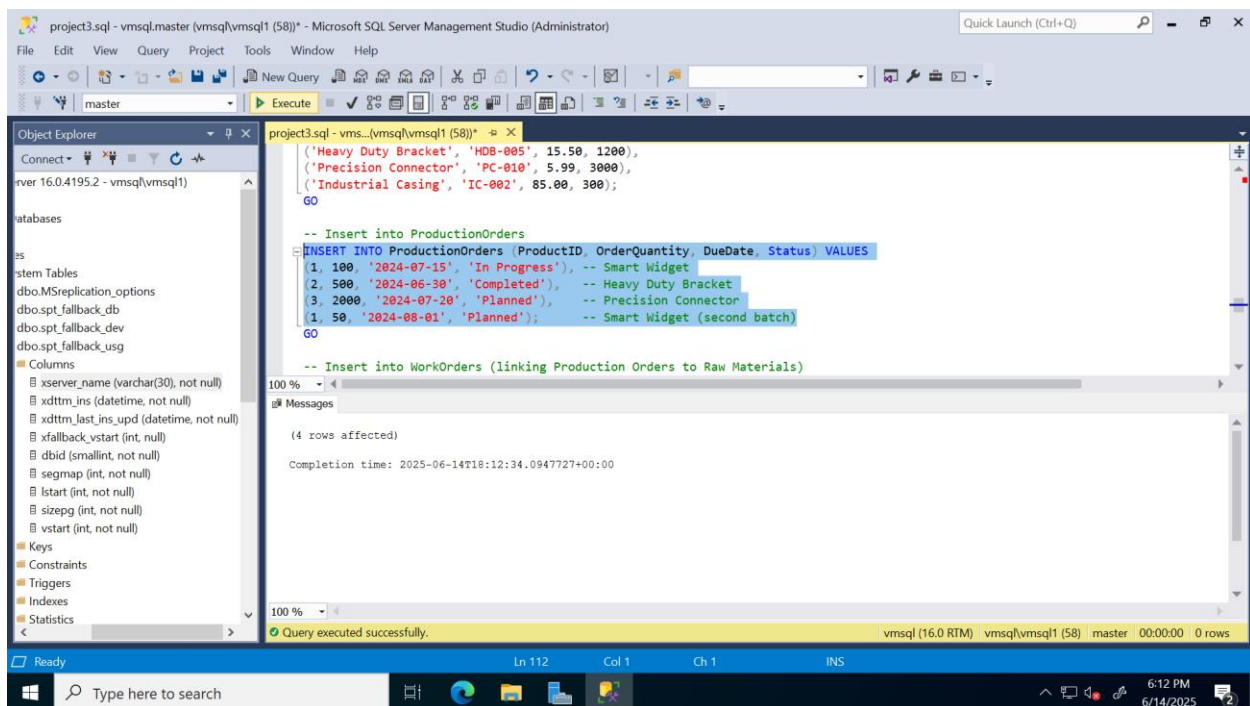
Executed select*from products all the details is executed successfully.



Executed create table production orders



Inserted value is orders it is successful



Executed the select statement of production order it is successful.

The screenshot shows the Microsoft SQL Server Management Studio (SSMS) interface. The query editor contains the following SQL statement:

```
-- SELECT * FROM ProductionOrders;
-- SELECT * FROM WorkOrders;
-- SELECT * FROM ManufacturingMetadata;
```

The query has been executed successfully, and the results are displayed in the Results pane. The results show a table with 4 rows and 6 columns: ProductionOrderID, ProductID, OrderQuantity, OrderDate, DueDate, and Status.

ProductionOrderID	ProductID	OrderQuantity	OrderDate	DueDate	Status
1	100	1	2025-06-14 18:12:34.073	2024-07-15	In Progress
2	2	500	2025-06-14 18:12:34.073	2024-06-30	Completed
3	3	2000	2025-06-14 18:12:34.073	2024-07-20	Planned
4	1	50	2025-06-14 18:12:34.073	2024-08-01	Planned

The status bar at the bottom indicates: Query executed successfully. vmsql (16.0 RTM) vmsql/vmsql1 (58) master 00:00:00 4 rows.

Executed the create statement of work order it is successful.

The screenshot shows the Microsoft SQL Server Management Studio (SSMS) interface. The query editor contains the following SQL statement:

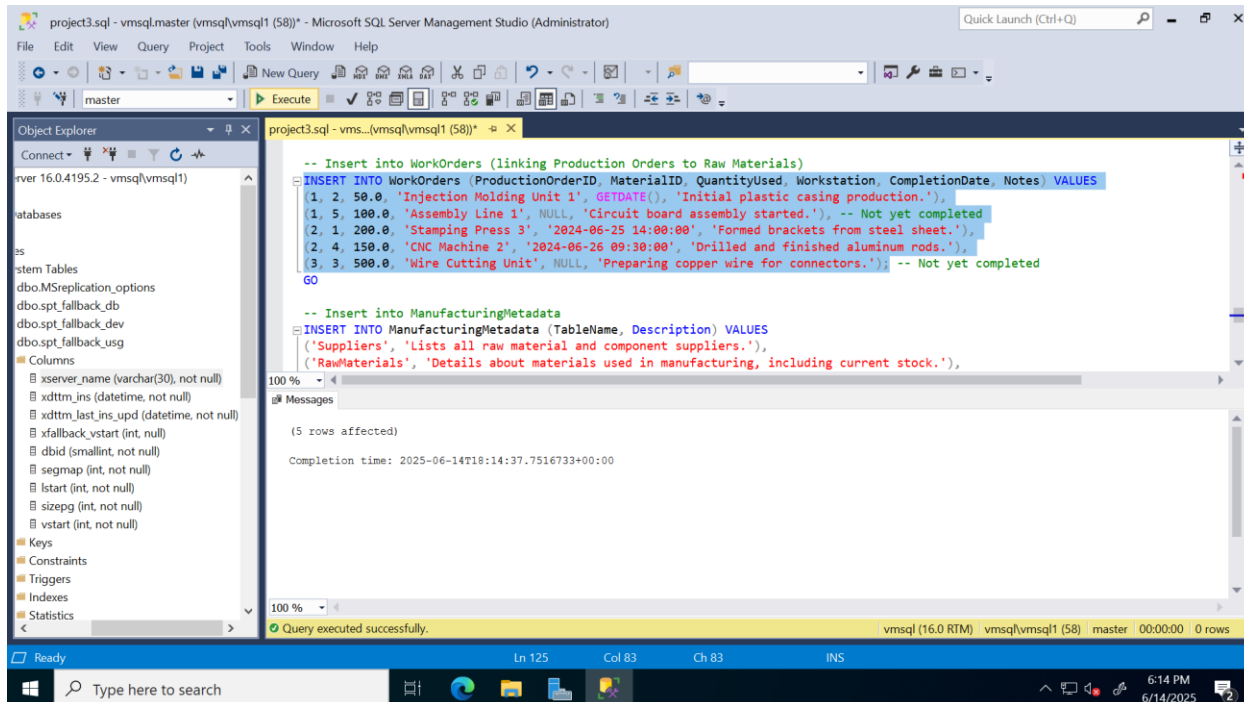
```
CREATE TABLE WorkOrders (
    WorkOrderID INT PRIMARY KEY IDENTITY(1,1),
    ProductionOrderID INT NOT NULL,
    MaterialID INT NOT NULL, -- Raw material used for this work step
    QuantityUsed DECIMAL(10, 2) NOT NULL,
    Workstation NVARCHAR(50),
    CompletionDate DATETIME,
    Notes NVARCHAR(500),
    FOREIGN KEY (ProductionOrderID) REFERENCES ProductionOrders(ProductionOrderID),
    FOREIGN KEY (MaterialID) REFERENCES RawMaterials(MaterialID)
);
```

The query has been executed successfully, and the Messages pane shows the following output:

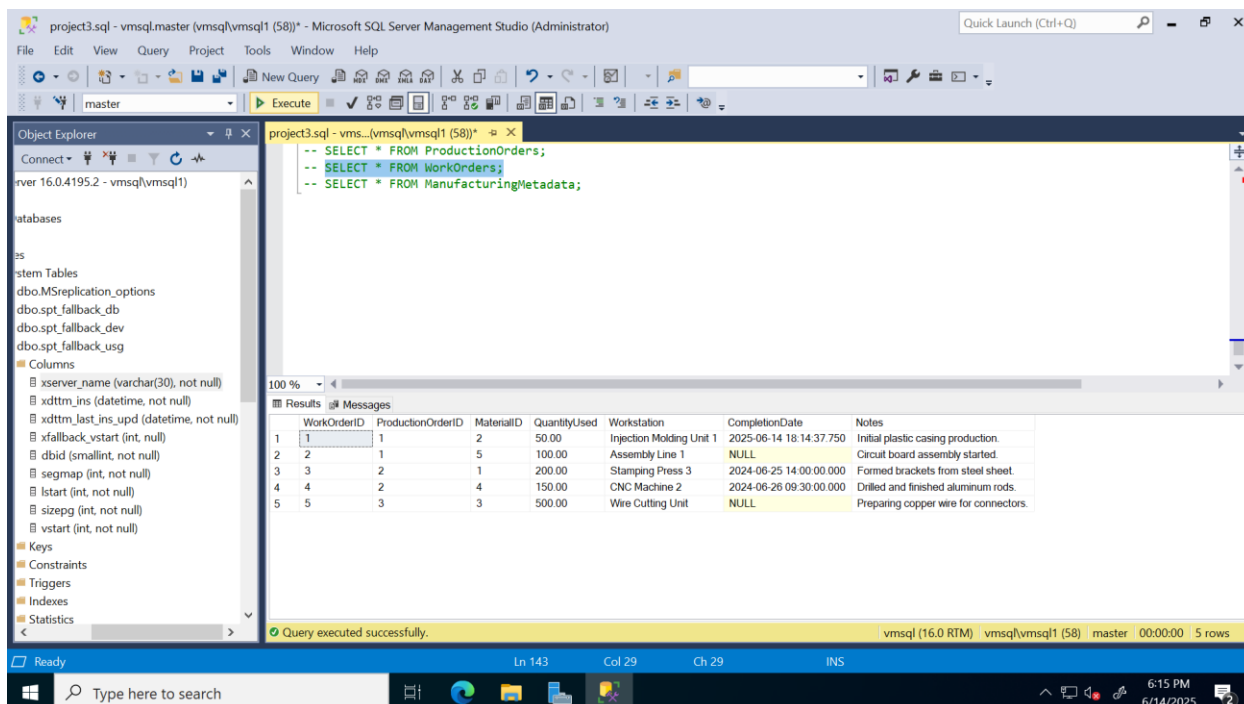
```
Commands completed successfully.
Completion time: 2025-06-14T18:14:10.1422948+00:00
```

The status bar at the bottom indicates: Query executed successfully. vmsql (16.0 RTM) vmsql/vmsql1 (58) master 00:00:00 0 rows.

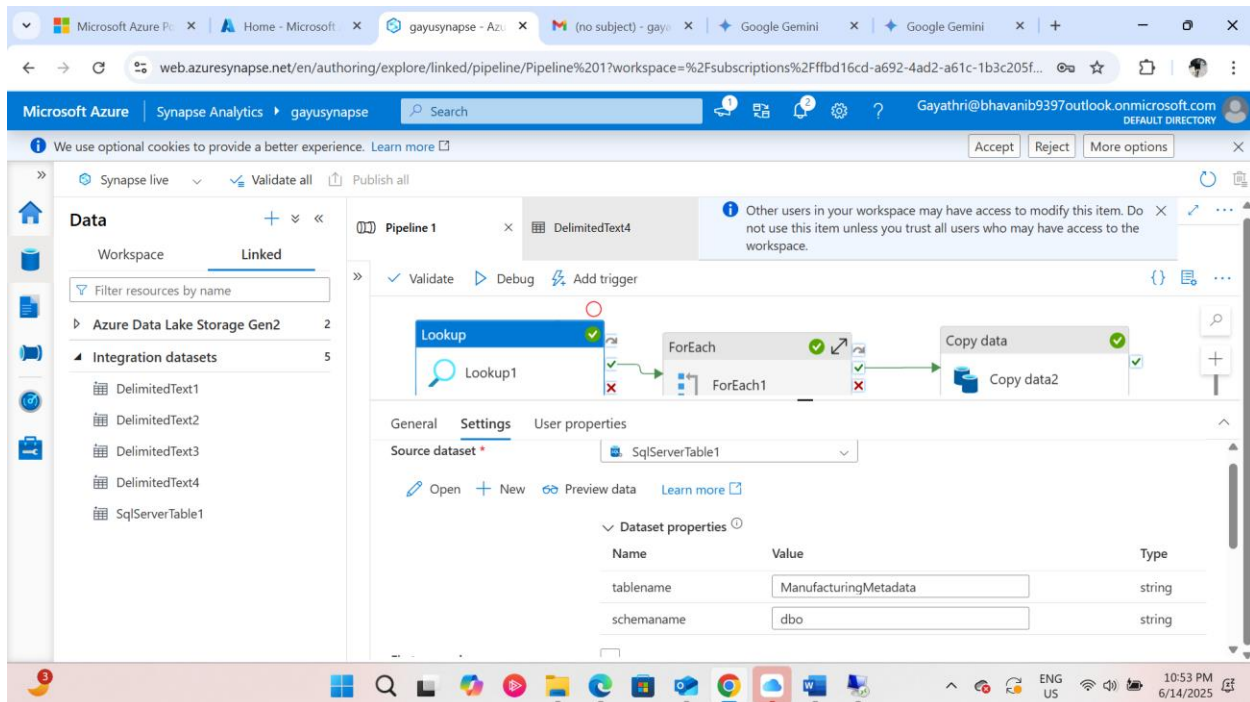
Inserted the statement of work order it is successful.



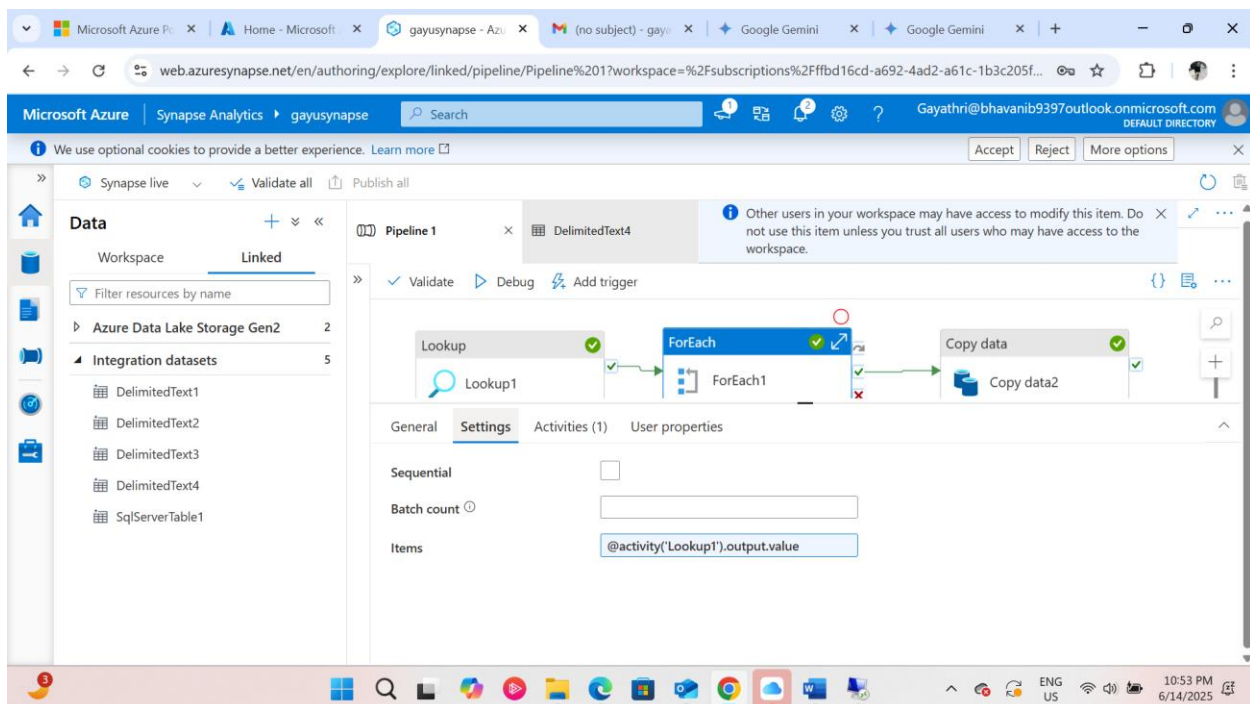
Executed the select statement of work order it is successful.



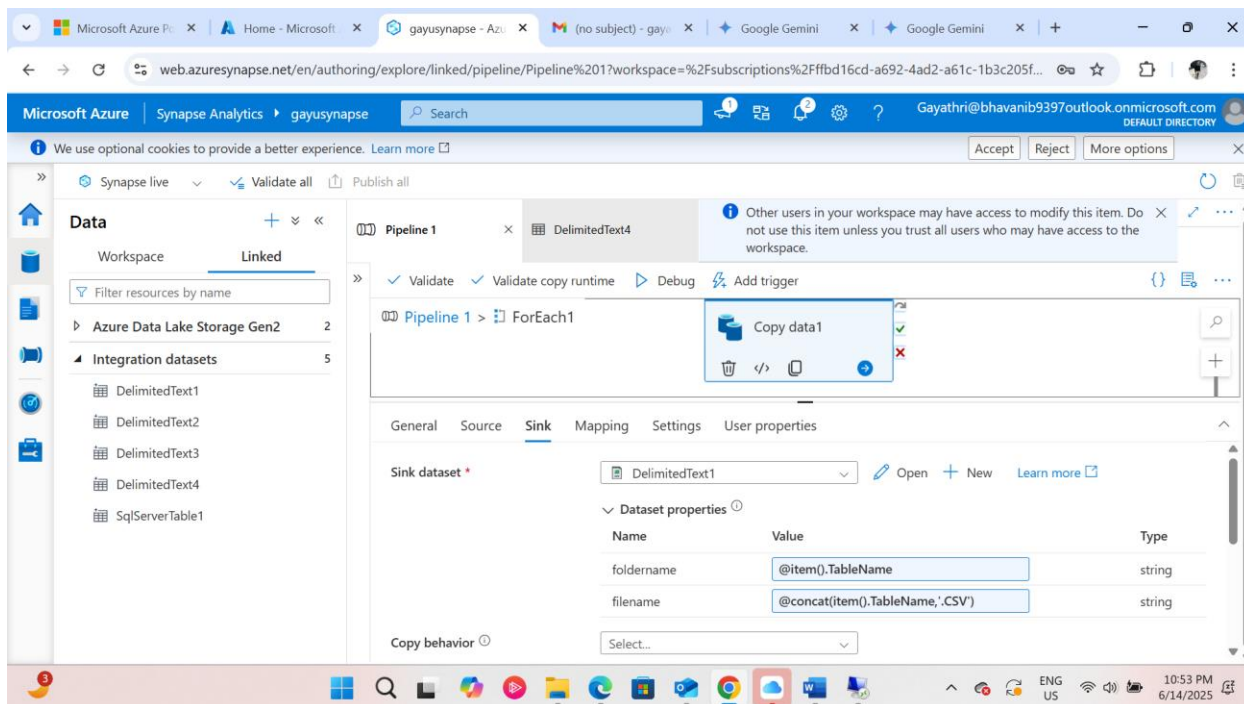
Open synapse, Inside the lookup I am giving source dataset as Sql server table 1 and giving table and dbo name in settings



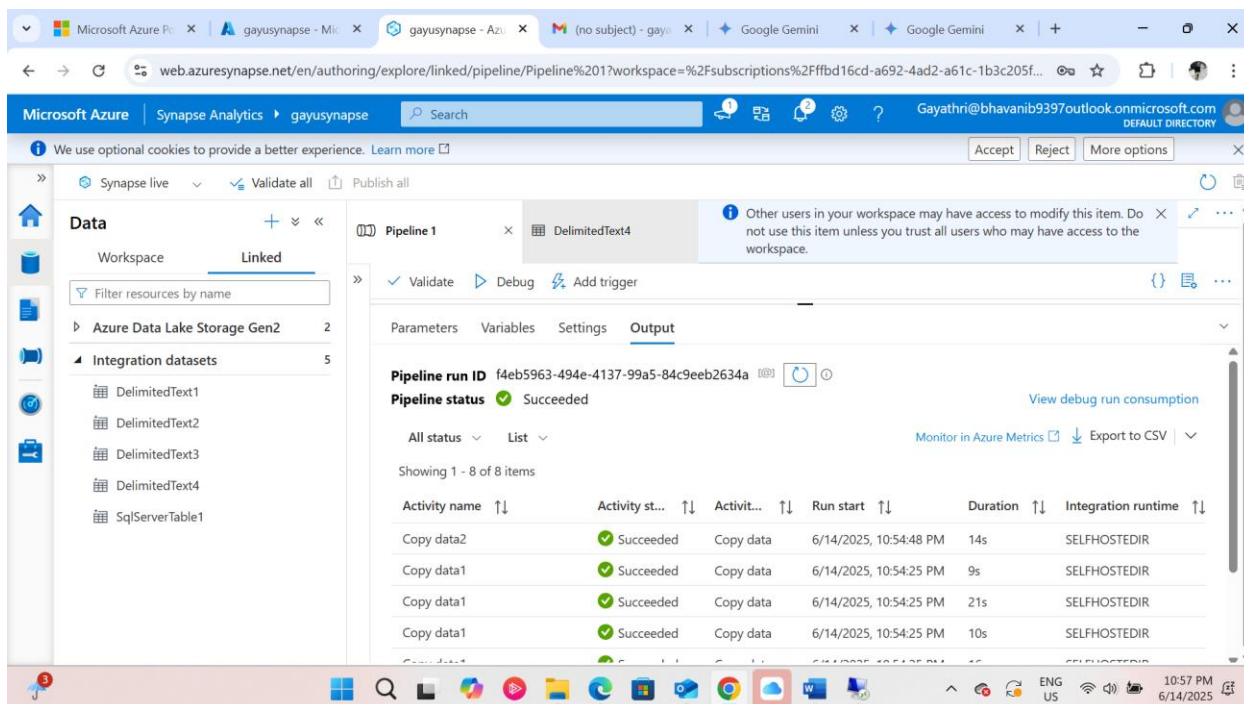
In settings in Item I am selecting Add dynamic content and choosing array value in For each



Inside Foreach I am selecting, copy data and in sink section I am giving foldername and filename and I am storing in synapse



Publish and debug the pipeline, it is successful



If I go inside the synapse, all 5 folder is there successfully.so on-premise to cloud is done successfully.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo and a search bar. The main content area displays the 'Overview' page for a Storage account named 'gayusynapse'. The page indicates the authentication method is 'Access key' and the location is 'gayusynapse'. A table lists five blobs: 'ProductionOrders', 'Products', 'RawMaterials', 'Suppliers', and 'WorkOrders', all created on 6/14/2025. The left sidebar shows navigation options like 'Overview', 'Diagnose and solve problems', 'Access Control (IAM)', and 'Settings'.

Now, I am clicking on production order CSV and download it

Click on Edit option and click on preview.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'gayusynapse' container is selected under 'Storage accounts'. The main pane shows the 'ProductionOrders/ProductionOrders.CSV' blob. The 'Edit' tab is active, displaying a CSV file with 12 rows of data. The 'Preview' button is highlighted.

ProductionOrderID	ProductID	OrderQuantity	OrderDate	DueDate	Status
4,1,100	2025-06-14	21:42:07.9400000	2024-07-15	00:00:00.0000000	"In Progress"
5,2,500	2025-06-14	21:42:07.9400000	2024-06-30	00:00:00.0000000	"Completed"
6,3,2000	2025-06-14	21:42:07.9400000	2024-07-20	00:00:00.0000000	"Planned"
7,1,50	2025-06-14	21:42:07.9400000	2024-08-01	00:00:00.0000000	"Planned"
8,1,100	2025-06-14	21:43:18.9230000	2024-07-15	00:00:00.0000000	"In Progress"
9,2,500	2025-06-14	21:43:18.9230000	2024-06-30	00:00:00.0000000	"Completed"
10,3,2000	2025-06-14	21:43:18.9230000	2024-07-20	00:00:00.0000000	"Planned"
11,1,50	2025-06-14	21:43:18.9230000	2024-08-01	00:00:00.0000000	"Planned"
12,1,100	2025-06-14	21:44:09.6230000	2024-07-15	00:00:00.0000000	"In Progress"
13,2,500	2025-06-14	21:44:09.6230000	2024-06-30	00:00:00.0000000	"Completed"
14,3,2000	2025-06-14	21:44:09.6230000	2024-07-20	00:00:00.0000000	"Planned"

One of the table production order is executed successfully.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'gayusynapse' container is selected under 'Storage accounts'. The main pane shows the 'ProductionOrders/ProductionOrders.CSV' blob. The 'Edit' tab is active, displaying a table view of the CSV data. The table has 10 rows of data.

ProductionOrderID	ProductID	OrderQuantity	OrderDate	DueDate	Status
4	1	100	2025-06-14 21:42:07.9400000	2024-07-15 00:00:00.0000000	In Progress
5	2	500	2025-06-14 21:42:07.9400000	2024-06-30 00:00:00.0000000	Completed
6	3	2000	2025-06-14 21:42:07.9400000	2024-07-20 00:00:00.0000000	Planned
7	1	50	2025-06-14 21:42:07.9400000	2024-08-01 00:00:00.0000000	Planned
8	1	100	2025-06-14 21:43:18.9230000	2024-07-15 00:00:00.0000000	In Progress
9	2	500	2025-06-14 21:43:18.9230000	2024-06-30 00:00:00.0000000	Completed
10	3	2000	2025-06-14 21:43:18.9230000	2024-07-20 00:00:00.0000000	Planned

Now, I am clicking on Products CSV

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the user profile 'Gayathri@bhanib939...'. The main content area displays the 'gayusynapse' container overview. The left sidebar shows the 'Overview' tab selected. The main area shows the 'Products' blob with a search bar and a table of blobs. The table has columns: Name, Modified, Access tier, Archive status, Blob type, and Size. The 'Products.CSV' blob is listed with a modified date of 6/14/2025, 10:54:32, an access tier of 'Hot (Inferred)', a blob type of 'Block blob', and a size of 253.

Name	Modified	Access tier	Archive status	Blob type	Size
[-]					
Products.CSV	6/14/2025, 10:54:32 ...	Hot (Inferred)		Block blob	253

Click Edit and click on preview

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the user profile 'Gayathri@bhanib939...'. The main content area displays the 'Products/Products.CSV' blob edit view. The left sidebar shows the 'Overview' tab selected. The main area shows the 'Products.CSV' blob with a search bar and a table of blobs. The table has columns: Name, Modified, Access tier, Archive status, Blob type, and Size. The 'Products.CSV' blob is listed with a modified date of 6/14/2025, 10:54:32, an access tier of 'Hot (Inferred)', a blob type of 'Block blob', and a size of 253.

The 'Edit' tab is selected, and the CSV content is displayed in a text editor. The content is as follows:

```

1 ProductID,ProductName,ProductCode,Description,SellingPrice,CurrentFGStock
2 1,"Smart Widget v1.0","SW-001",,129.99,500
3 2,"Heavy Duty Bracket","HDB-005",,15.50,1200
4 3,"Precision Connector","PC-010",,5.99,3000
5 4,"Industrial Casing","IC-002",,85.00,300
6

```

The 'Preview' button is visible at the bottom of the text editor.

One of the table, Product is successfully viewed.

Microsoft Azure portal interface showing the 'Products/Products.CSV' blob. The left sidebar lists navigation options: Overview, Diagnose and solve problems, Access Control (IAM), Settings, Shared access tokens, Manage ACL, Access policy, Properties, and Metadata. The main content area displays the blob's details, including authentication method (Access key), location (gayusynapse / Products), and a table of product data.

ProductID	ProductName	ProductCode	Description	SellingPrice	CurrentFGStock
1	Smart Widget v1.0	SW-001		129.99	500
2	Heavy Duty Bracket	HDB-005		15.50	1200
3	Precision Connector	PC-010		5.99	3000
4	Industrial Casing	IC-002		85.00	300

Now, we are clicking on suppliers.

Microsoft Azure portal interface showing the 'Suppliers.CSV' blob. The left sidebar lists navigation options: Overview, Diagnose and solve problems, Access Control (IAM), Settings, Shared access tokens, Manage ACL, Access policy, Properties, and Metadata. The main content area displays the blob's details, including authentication method (Access key), location (gayusynapse / Suppliers), and a table of supplier data.

Name	Modified	Access tier	Archive status	Blob type	Size
[.]					
Suppliers.CSV	6/14/2025, 10:54:38 ...	Hot (Inferred)		Block blob	383

Click on edit and preview, it is viewed successfully.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'gayusynapse' container is selected, showing a list of blobs including 'RawMaterials.CSV'. The main pane displays the 'RawMaterials/RawMaterials.CSV' blob. The 'Edit' tab is active, showing a CSV file with the following data:

MaterialID	MaterialName	Description	UnitOfMeasure	CurrentStock	SupplierID
1	Stainless Steel Sheet		kg	5000	1
2	ABS Plastic Pellets		kg	2500	2
3	Copper Wire (1mm)		meters	10000	3
4	Aluminum Rod		meters	3000	1
5	Circuit Board (Empty)		pieces	1500	3

At the bottom of the editor, there is a 'Preview' button.

Now, In Raw material folder click on Edit and preview, table is viewed successfully.

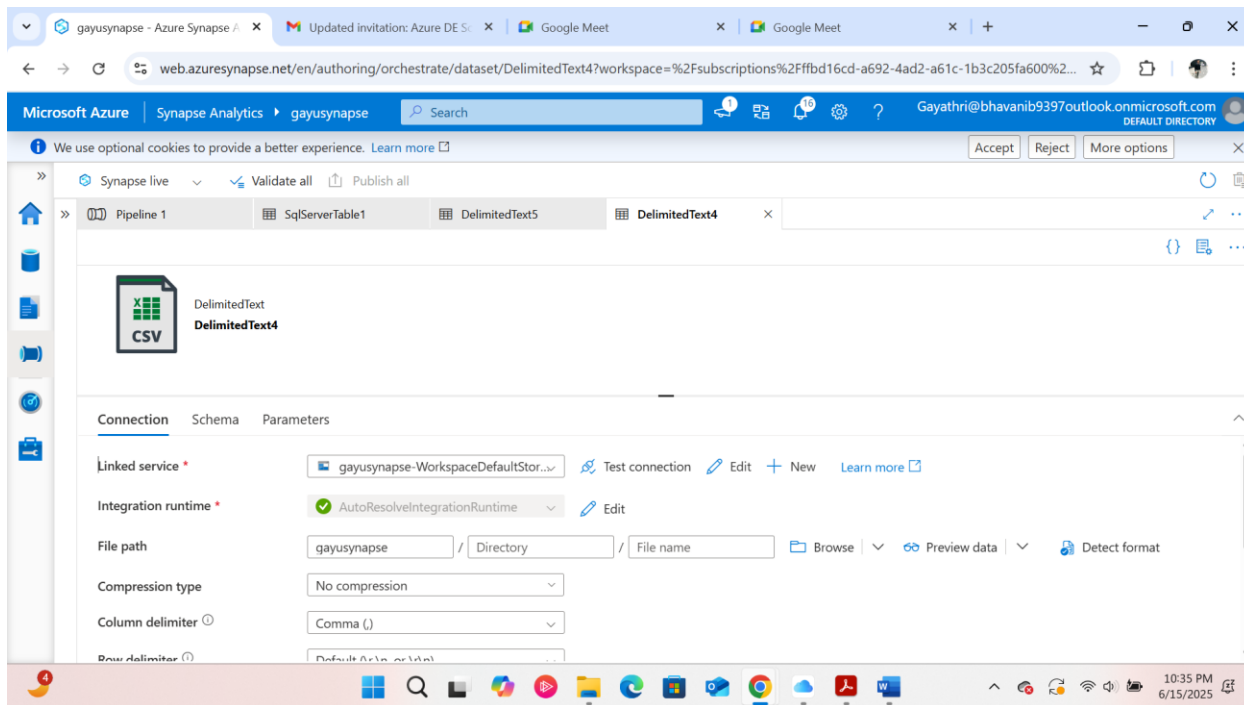
The screenshot shows the Microsoft Azure portal interface. On the left, the 'gayusynapse' container is selected, showing a list of blobs including 'RawMaterials.CSV'. The main pane displays the 'RawMaterials/RawMaterials.CSV' blob. The 'Edit' tab is active, showing a table view of the CSV data:

MaterialID	MaterialName	Description	UnitOfMeasure	CurrentStock	SupplierID
1	Stainless Steel Sheet		kg	5000	1
2	ABS Plastic Pellets		kg	2500	2
3	Copper Wire (1mm)		meters	10000	3
4	Aluminum Rod		meters	3000	1
5	Circuit Board (Empty)		pieces	1500	3

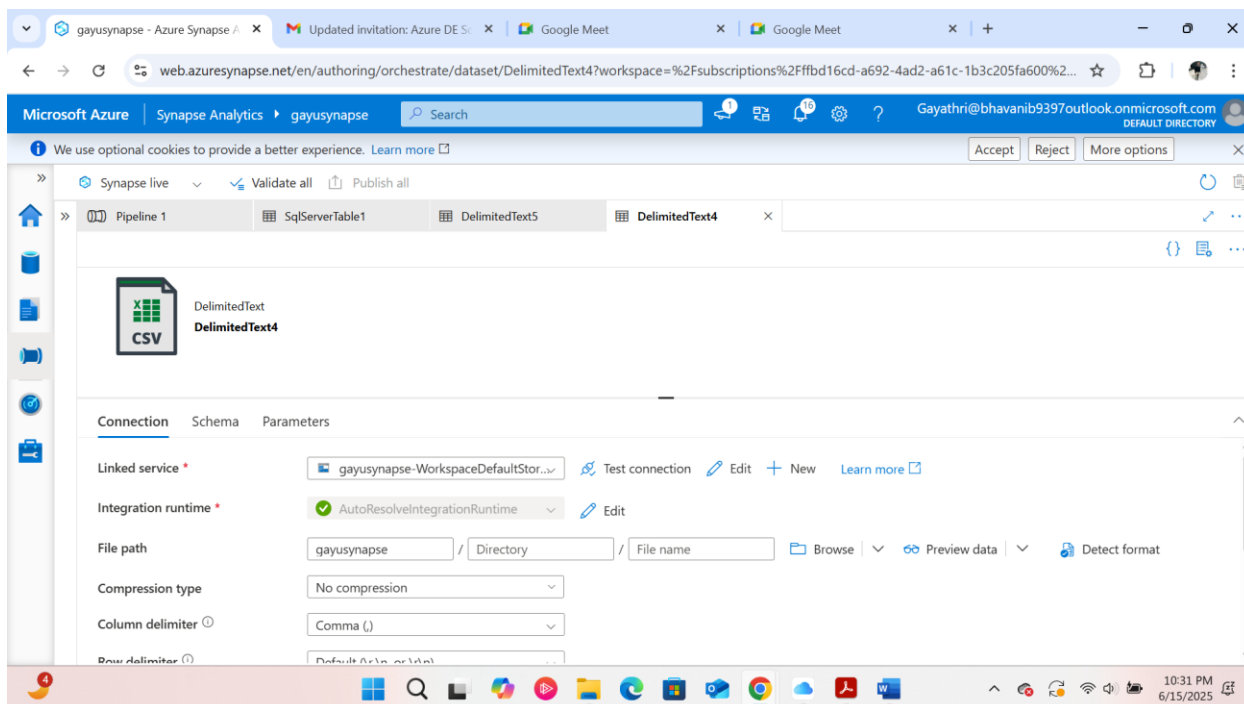
At the bottom of the editor, there is an 'Edit' button.

Now, I am taking another copy data to copy from cloud to on-premise

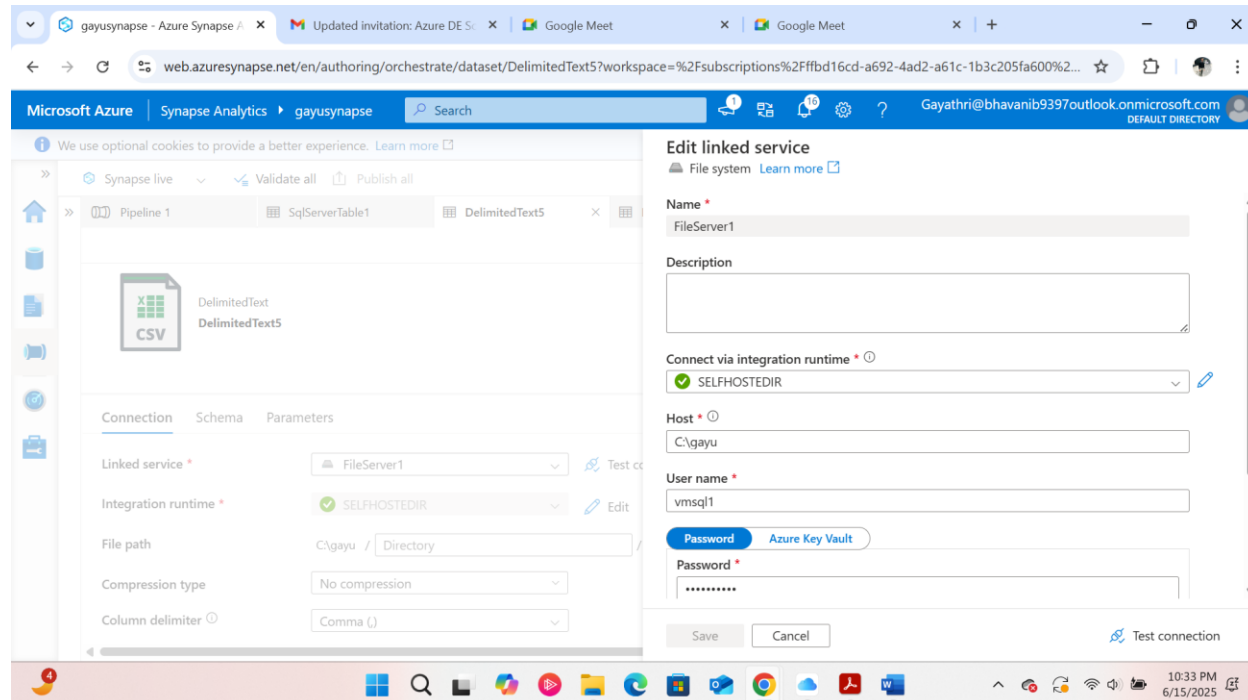
On source select synapse workspace from cloud and click on source file path



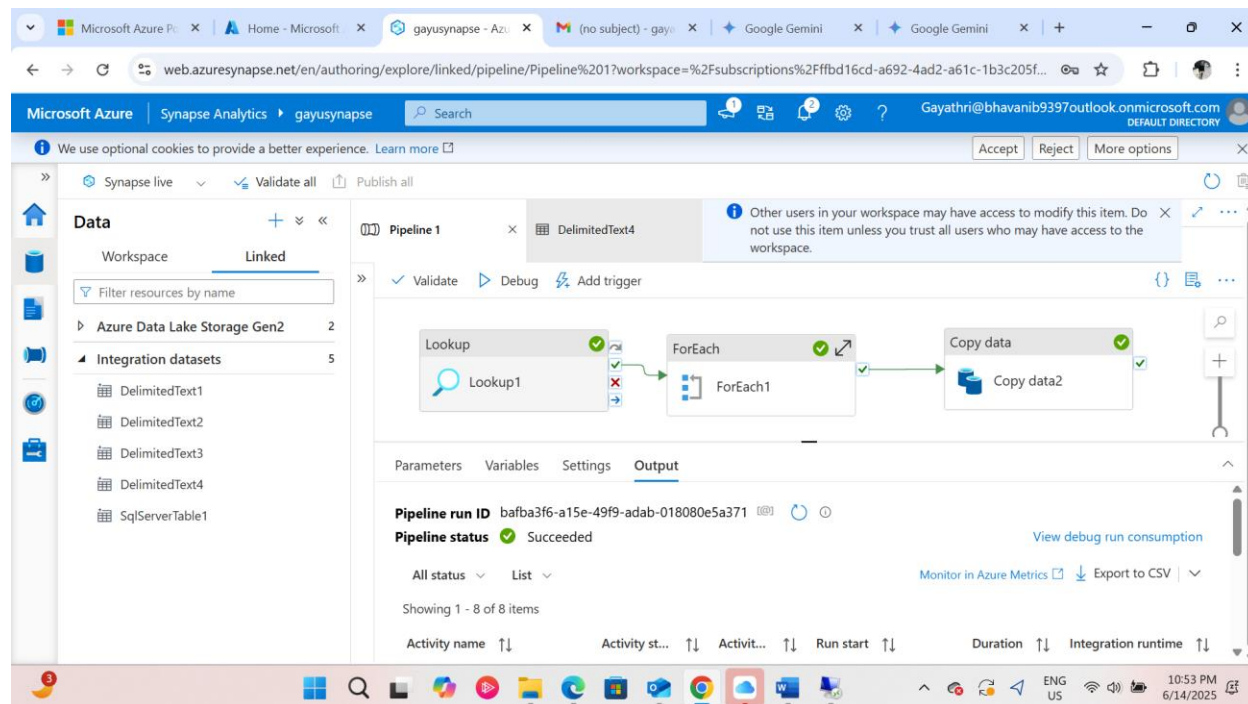
In sink path say the location where it should be saved.



Linked service for sink should be fileserver1 and click on save



Publish and debug the pipeline



Debug is successful.

The screenshot displays the Microsoft Azure Synapse Analytics web portal. The left sidebar shows the 'Data' section with 'Workspace' and 'Linked' tabs. Under 'Linked', there are five integration datasets: DelimitedText1, DelimitedText2, DelimitedText3, DelimitedText4, and SqlServerTable1. The main area shows 'Pipeline 1' with a visual workflow diagram. The workflow consists of three steps: 'Lookup' (Lookup1), 'ForEach' (ForEach1), and 'Copy data' (Copy data2). All steps are marked with green checkmarks, indicating successful execution. Below the diagram, the 'Output' tab is selected, showing a table of execution results.

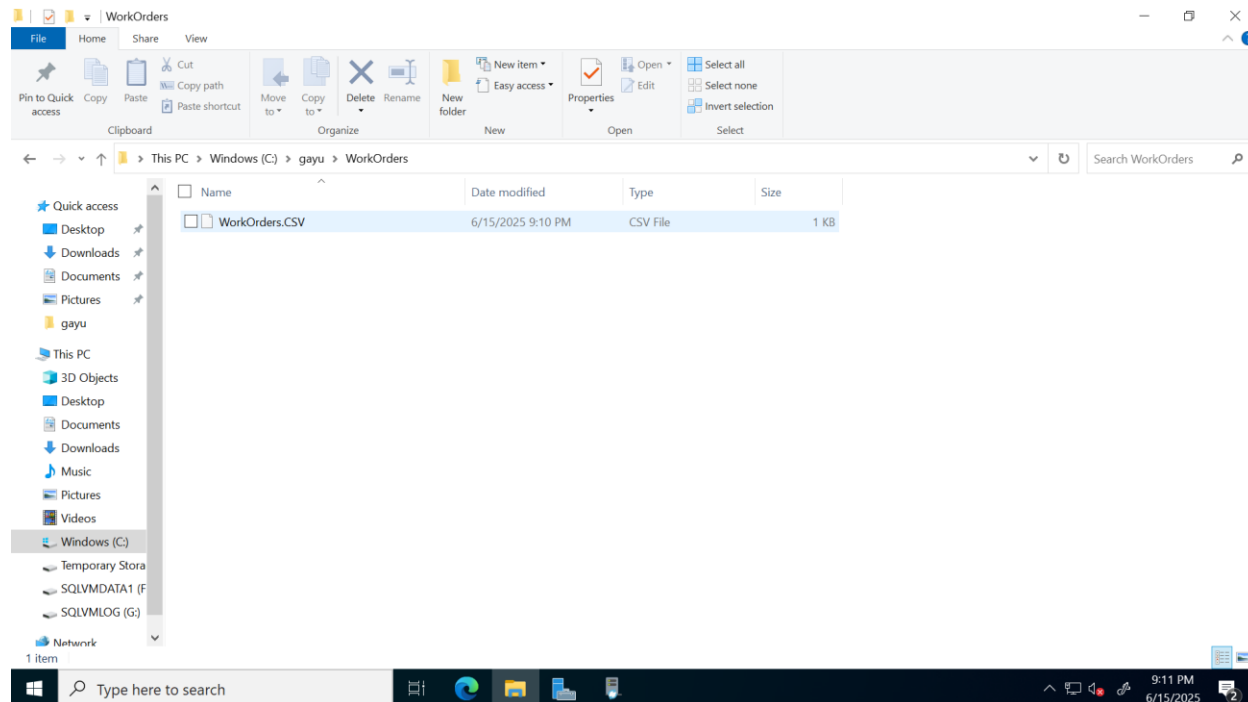
Copy data1	Variables	Settings	Output
Copy data1	✓ Succeeded	Copy data	6/14/2025, 11:14:58 PM 20s SELFHOSTEDIR
Copy data1	✓ Succeeded	Copy data	6/14/2025, 11:14:58 PM 15s SELFHOSTEDIR
Copy data1	✓ Succeeded	Copy data	6/14/2025, 11:14:58 PM 9s SELFHOSTEDIR
Copy data1	✓ Succeeded	Copy data	6/14/2025, 11:14:58 PM 14s SELFHOSTEDIR
Copy data1	✓ Succeeded	Copy data	6/14/2025, 11:14:58 PM 10s SELFHOSTEDIR
ForEach1	✓ Succeeded	ForEach	6/14/2025, 11:14:57 PM 25s
Lookup1	✓ Succeeded	Lookup	6/14/2025, 11:14:48 PM 9s SELFHOSTEDIR

Now, go to virtual Machine, click on c driver and click on gayu folder, the 5 folder is seen successfully.

The screenshot shows a Windows File Explorer window titled 'gayu'. The address bar indicates the path is 'This PC > Windows (C:) > gayu'. The left sidebar shows the 'gayu' folder selected. The main pane displays a list of five folders, all created on 6/15/2025 at 9:10 PM. The folders are: WorkOrders, Suppliers, RawMaterials, Products, and ProductionOrders.

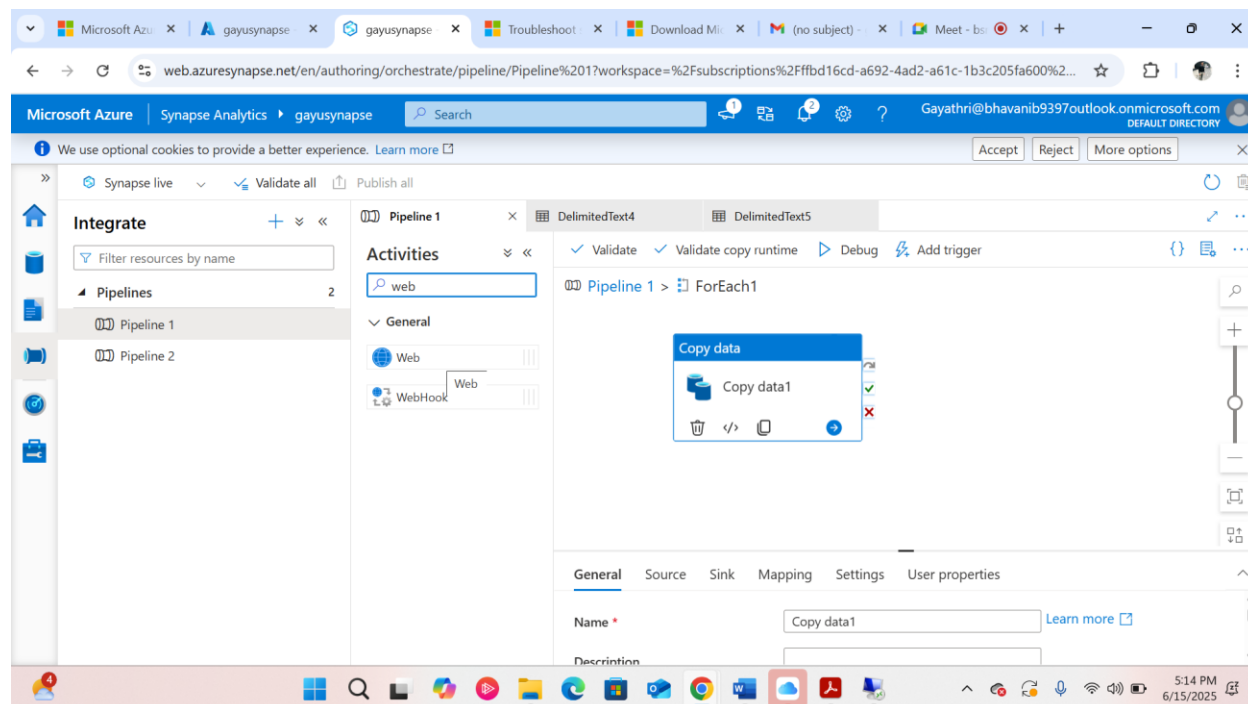
Name	Date modified	Type	Size
WorkOrders	6/15/2025 9:10 PM	File folder	
Suppliers	6/15/2025 9:10 PM	File folder	
RawMaterials	6/15/2025 9:10 PM	File folder	
Products	6/15/2025 9:10 PM	File folder	
ProductionOrders	6/15/2025 9:10 PM	File folder	

Inside workorder folder, CSV file is seen successfully.

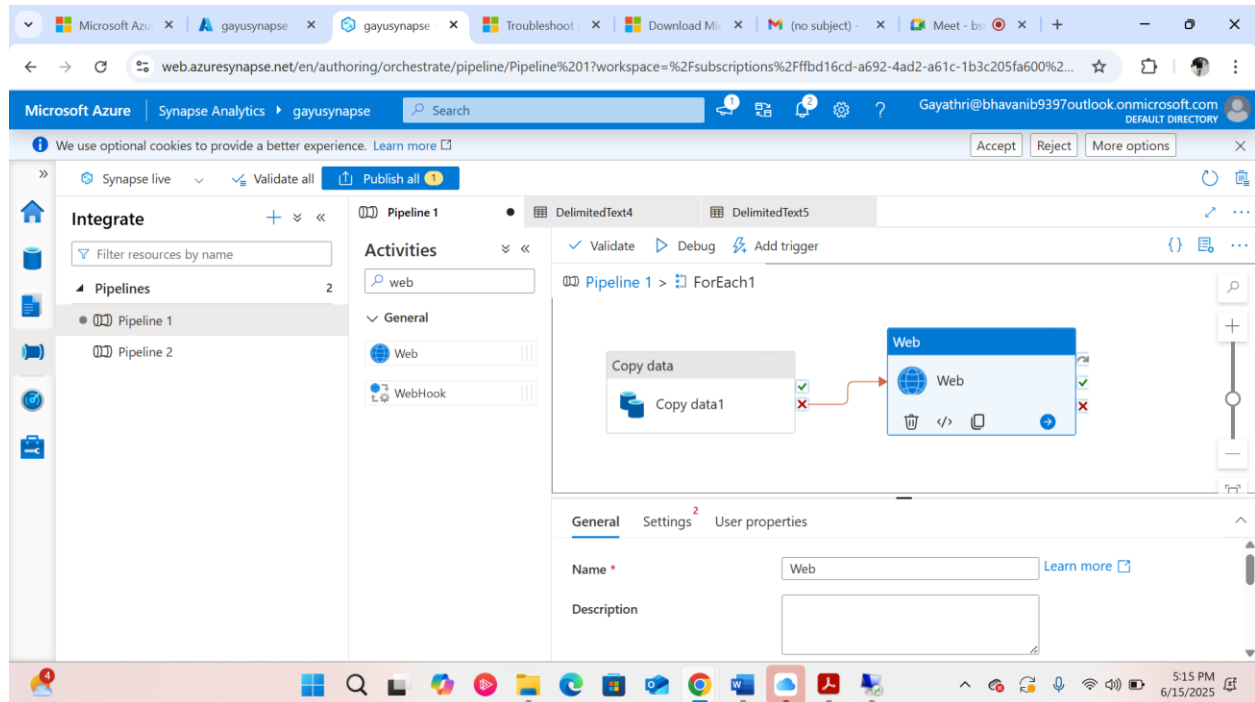


EMAIL ACTIVITY:

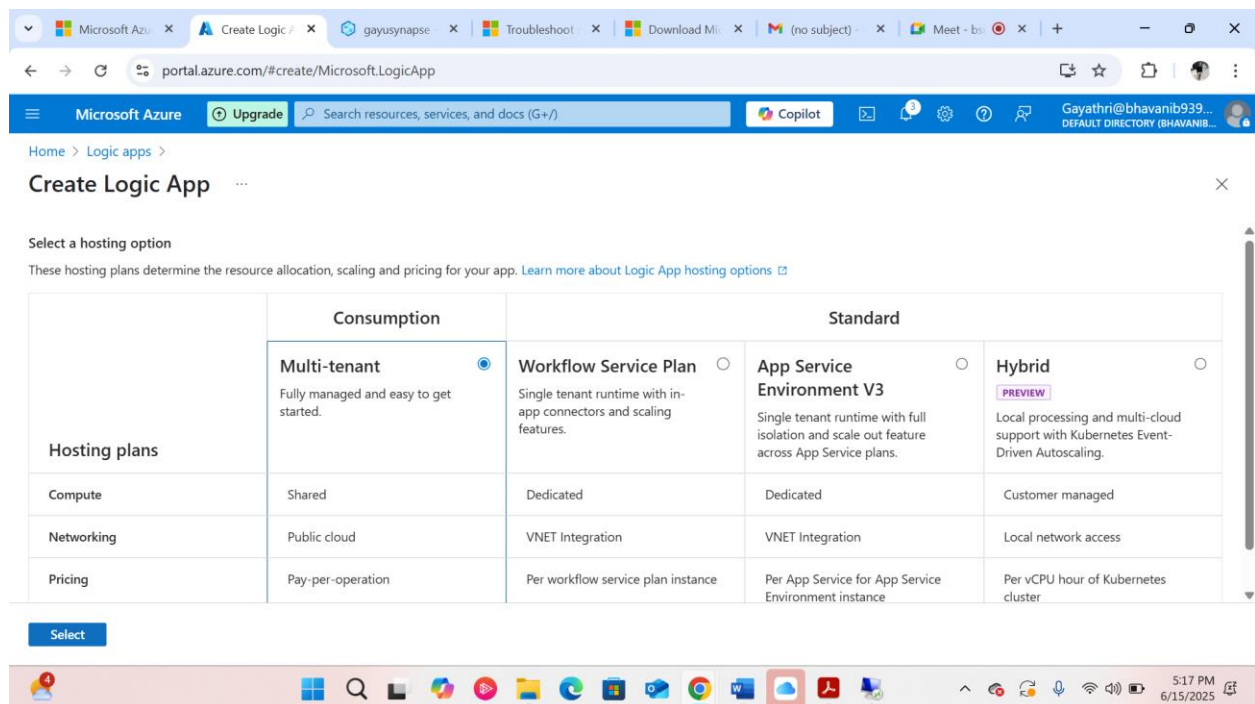
For email activity, I am selecting web from synapse workspace



Inside foreach, we already have copy data, now I am connecting copy data with web



Now, I am selecting Multi-tenant and click on select



I am creating Logic app, Selecting Region as West US, LOGIC APP NAME and click on review and create

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Logic apps > Create Logic App >

Create Logic App (Multi-tenant)

Subscription * Azure subscription 1

Resource Group * gayuadlsproje

Instance Details

Logic App name * gayulogic

Region * West US

Enable log analytics * ☐ Yes ☒ No

⚠ There are no log analytics workspace resources in the selected subscription. In order to enable log analytics, either create a new log analytics workspace resource or switch to a subscription which already has one.

Review + create | < Previous | Next: Tags >

Now, click on create

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Logic apps > Create Logic App >

Create Logic App (Multi-tenant)

Basics | Tags | **Review + create**

Summary

Logic App
by Microsoft

Details

Subscription	ffb16cd-a692-4ad2-a61c-1b3c205fa600
Resource Group	gayuadlsproje
Name	gayulogic
Region	westus
Log analytics	Disabled

Create | < Previous | Next > | Download a template for automation

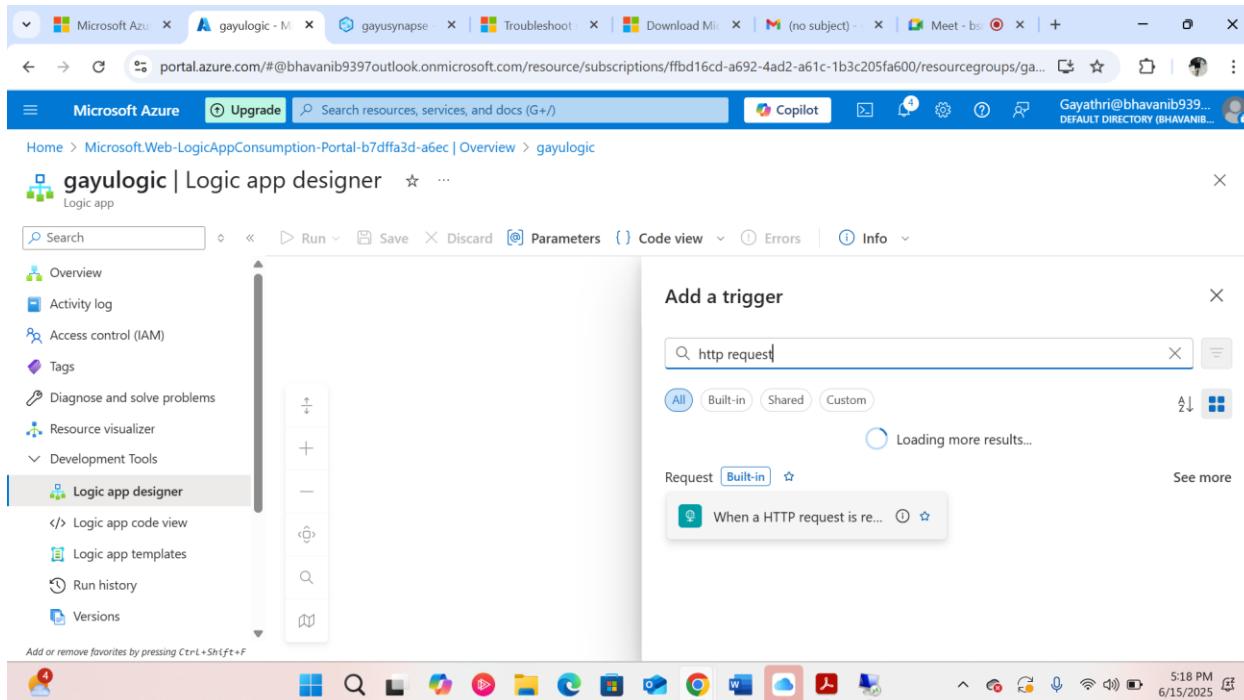
Deployment is successful, now go to resource

The screenshot shows the Microsoft Azure portal interface. The browser address bar displays the URL: `portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2Fffbd16cd-a692-4ad2-a61c-1b3c205fa600%2Fresourcegroups/gayualdsproje/providers/Microsoft.Logic/workflows/gayulogic`. The page title is "Microsoft.Web-LogicAppConsumption-Portal-b7dffa3d-a6ec | Overview". The left sidebar shows the "Overview" tab selected. The main content area displays a green checkmark and the message "Your deployment is complete". Below this, deployment details are shown: Deployment name: Microsoft.Web-LogicAppConsumption-Portal-b7dffa3d-a6ec, Start time: 6/15/2025, 5:17:54 PM, Subscription: Azure subscription 1, Correlation ID: 0fdd6560-4617-4500-b50a-9ce..., Resource group: gayualdsproje. A "Go to resource" button is visible. On the right, there are several informational cards: "Deployment in progress..." (Deployment to resource group 'gayualdsproje' is in progress), "Cost management" (Get notified to stay within your budget and prevent unexpected charges on your bill. Set up cost alerts >), "Microsoft Defender for Cloud" (Secure your apps and infrastructure. Go to Microsoft Defender for Cloud >), "Free Microsoft tutorials" (Start learning today >), and "Work with an expert". The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 5:18 PM on 6/15/2025.

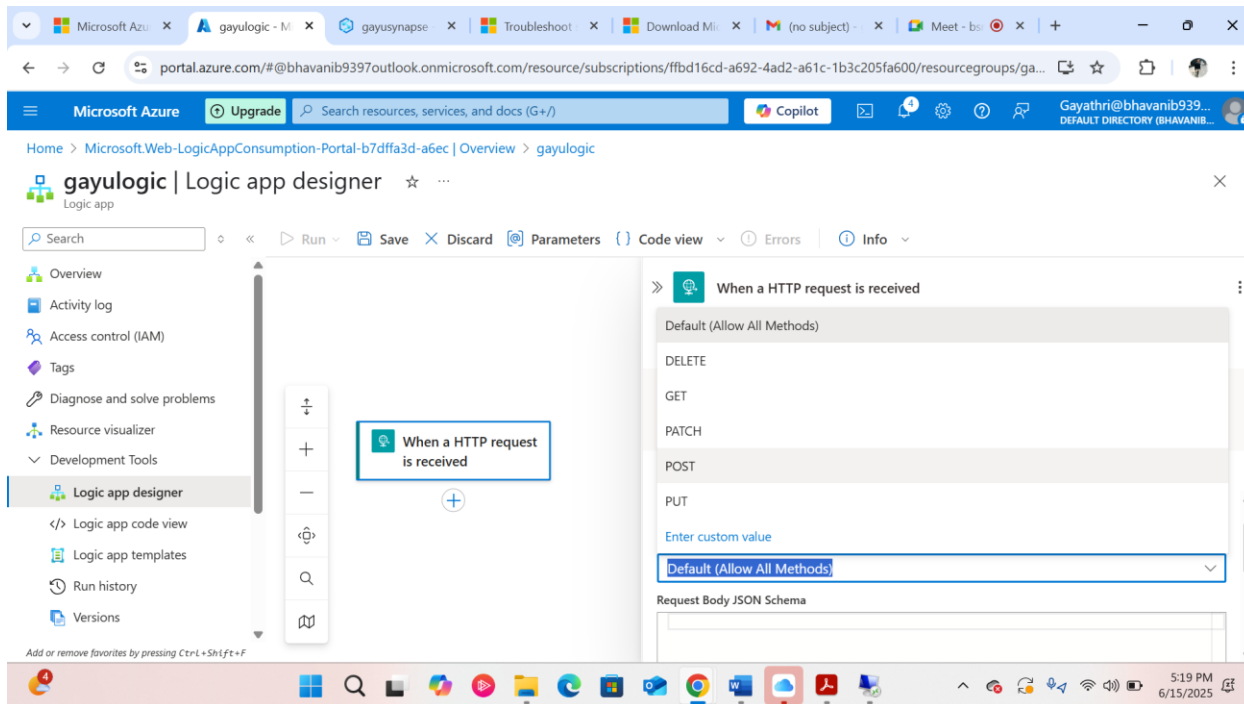
Click on LOGIC App designer under Development Tools

The screenshot shows the Microsoft Azure portal interface for the Logic App designer. The browser address bar displays the URL: `portal.azure.com/#@bhavanib9397outlook.onmicrosoft.com/resource/subscriptions/ffbd16cd-a692-4ad2-a61c-1b3c205fa600/resourcegroups/gayualdsproje/providers/Microsoft.Logic/workflows/gayulogic/designer`. The page title is "gayulogic | Logic app". The left sidebar shows the "Overview" tab selected. The main content area displays the Logic App designer interface. The "Essentials" section shows the following details: Resource group (move) gayualdsproje, Location (move) West US, Subscription (move) Azure subscription 1, Subscription ID ffb16cd-a692-4ad2-a61c-1b3c205fa600, Workflow URL --, Tags (edit) Add tags. The "Definition" section shows: 0 triggers, 0 actions, Status Enabled, Runs last 24 hours, 0 successful, 0 failed, Integration Account --. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 5:18 PM on 6/15/2025.

Type Http request and click ok



Click on POST



Click inside body Json schema

Microsoft Azure | Logic app designer

When a HTTP request is received

Changing the trigger name updates the callback URL when you save the workflow.

Add a description

Parameters Settings Code view About

Request Body JSON Schema

Use sample payload to generate schema

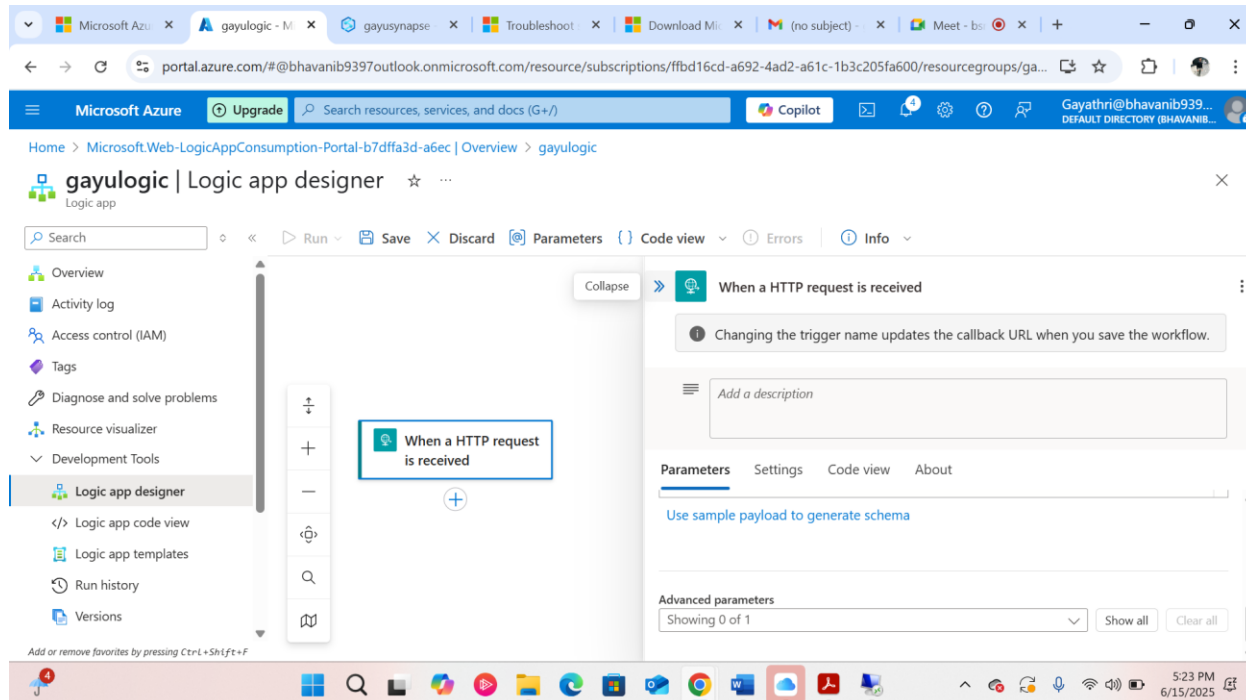
Type the sample Json Payload and click on Done

Enter or paste a sample JSON payload.

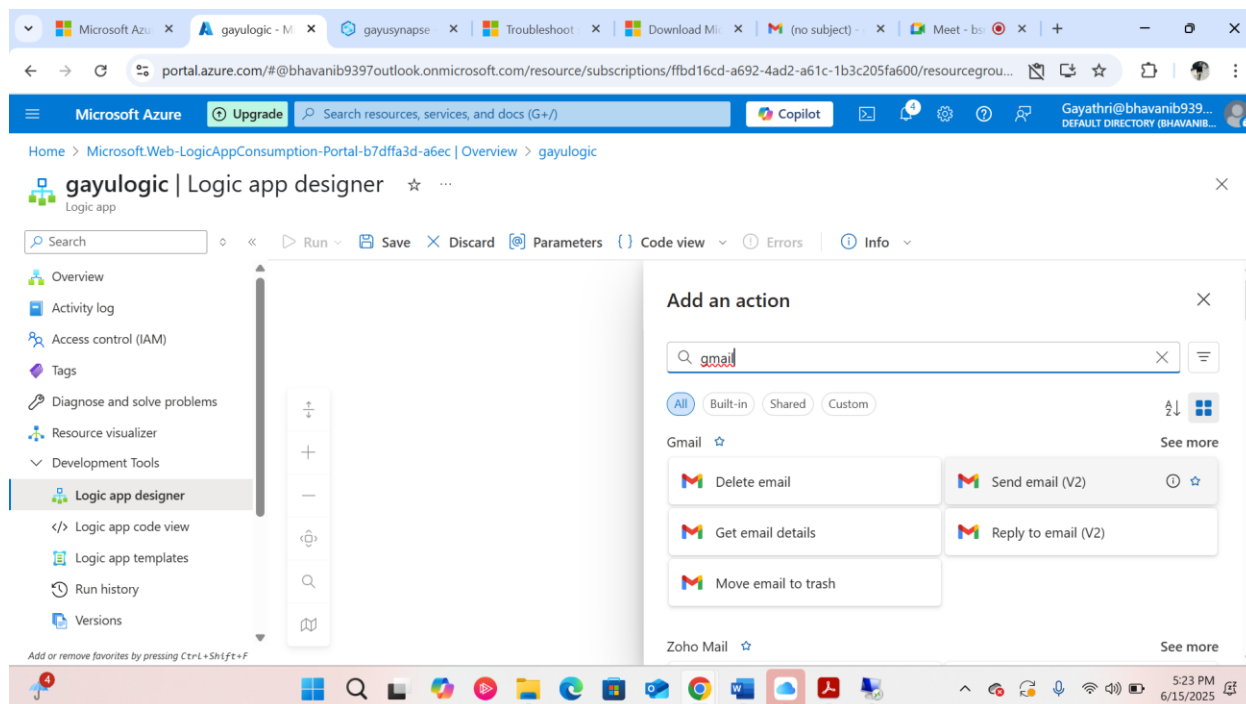
```
1 {
2   "PipelineName" : "Test",
3   "ErrorMessage" : "test",
4   "WorkspaceName" : "test"
5 }
```

Done

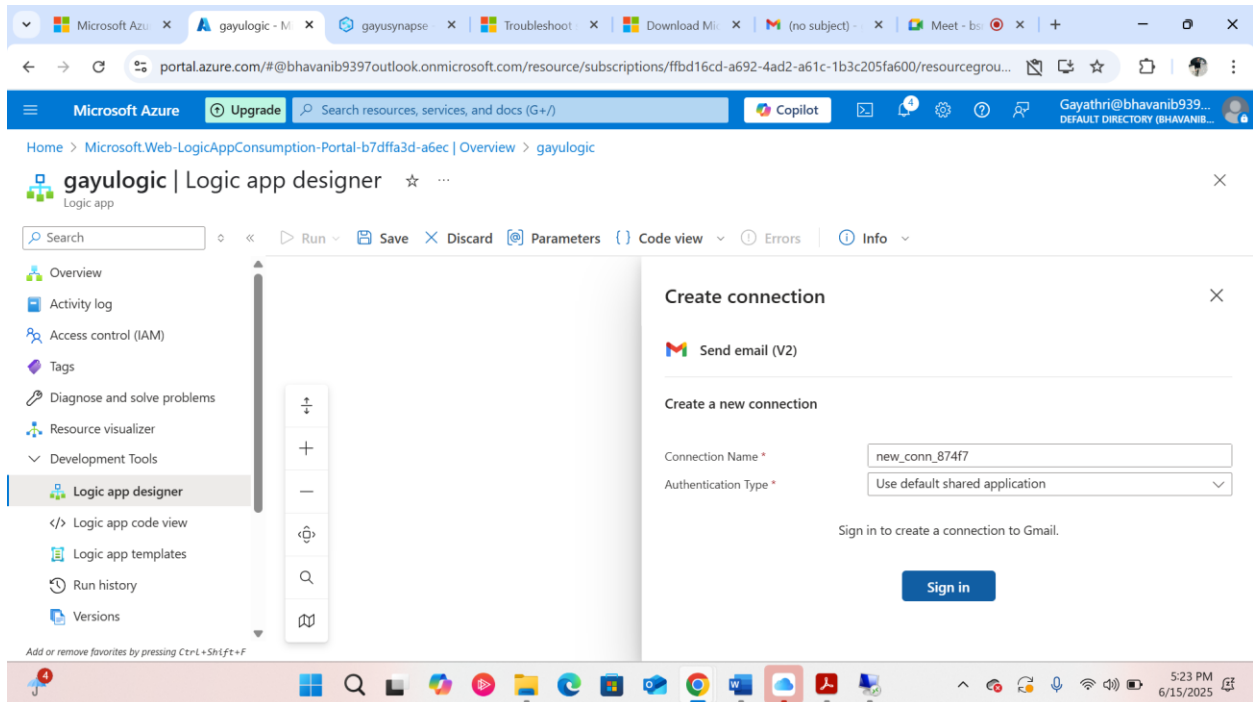
Now, click on Collapse



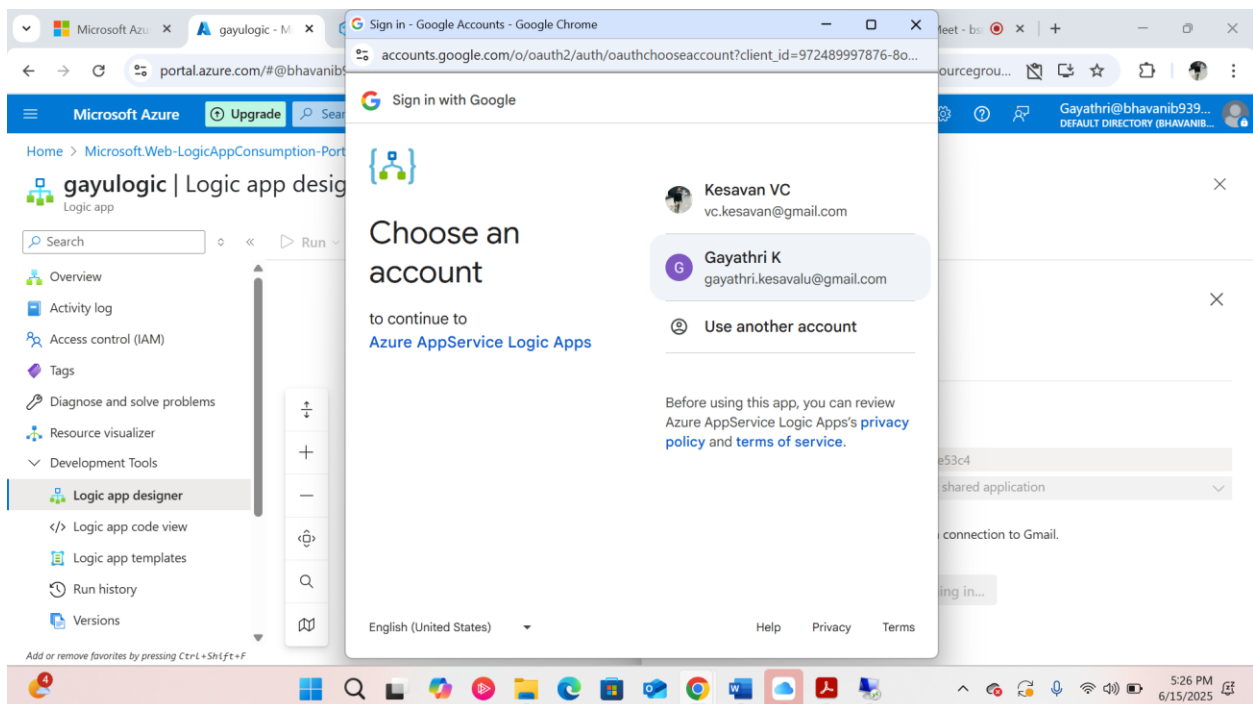
Click on +sign and type gmail in Add an action



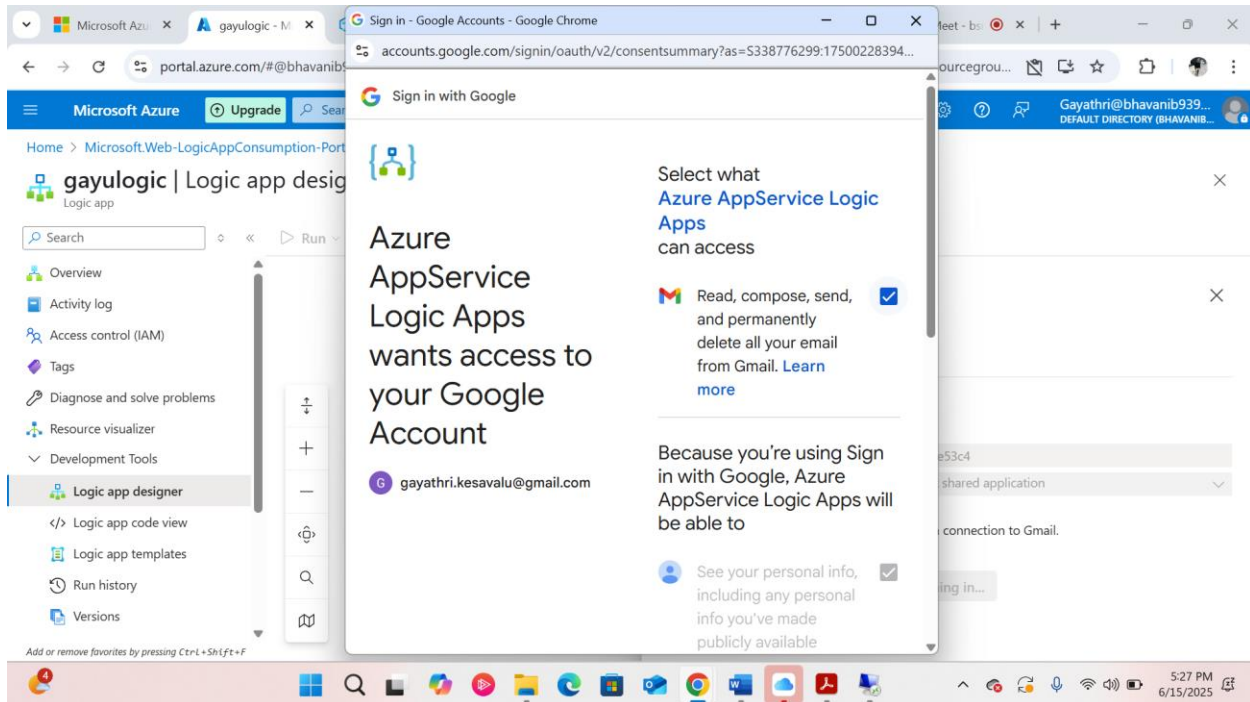
Click on Sign in



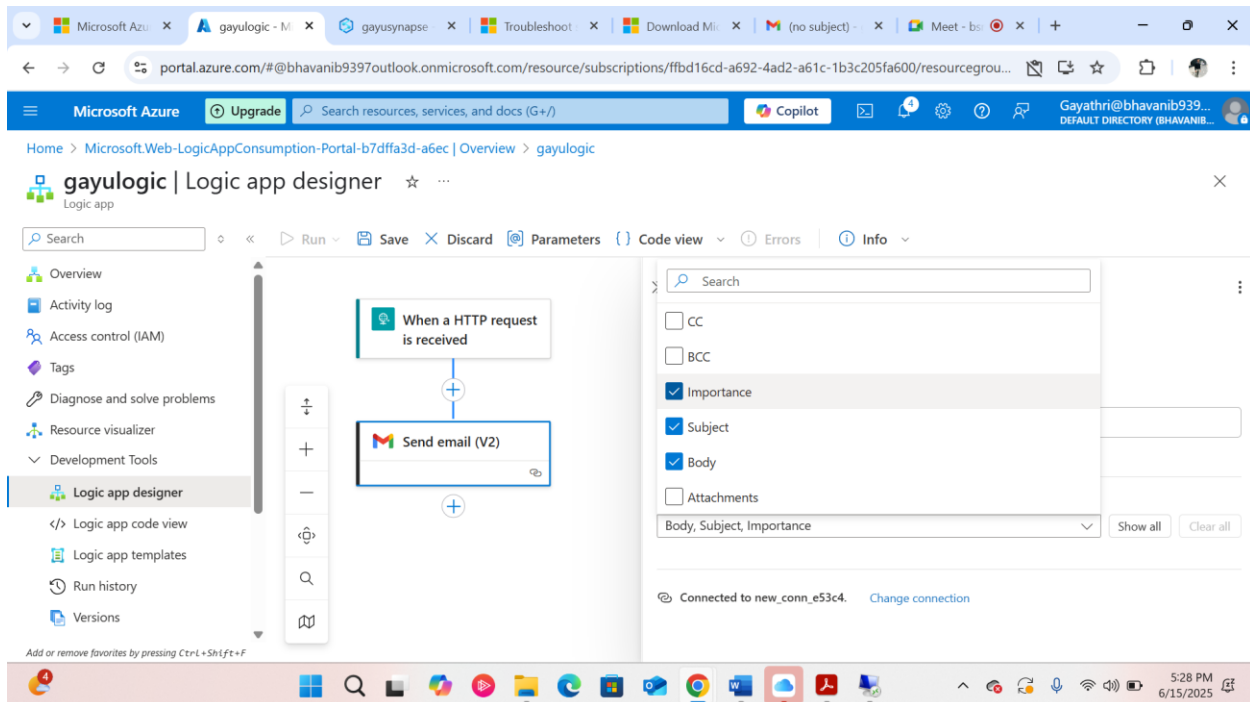
Choose the email



Select Read, compose, Send option



Select Important, Body, subject



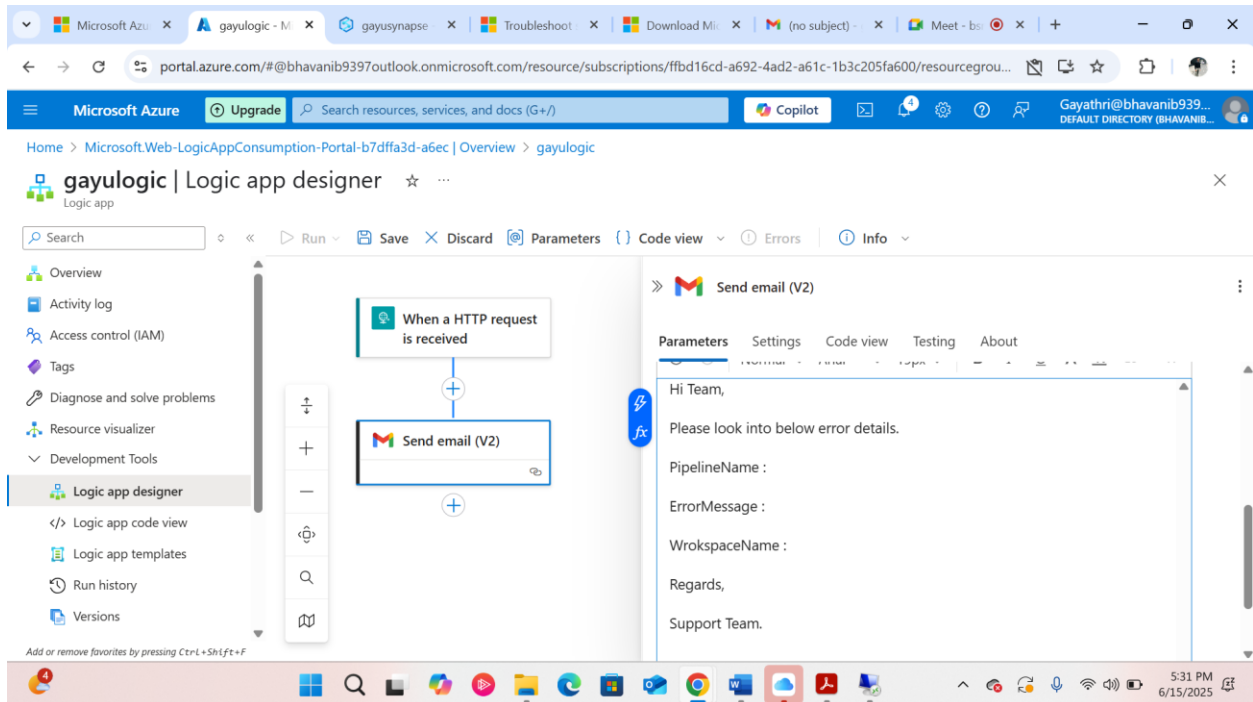
Type To email address

The screenshot shows the Microsoft Azure Logic App Designer interface. The left sidebar contains navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Development Tools, and Logic app designer (selected). The Logic app designer section includes options for Logic app code view, Logic app templates, Run history, and Versions. The main workspace displays a logic flow starting with 'When a HTTP request is received' followed by 'Send email (V2)'. The right-hand pane shows the configuration for the 'Send email (V2)' action. The 'Parameters' tab is active, showing the 'To' field with the email address 'gayathri.kesavalu@gmail.com,vc.kesavan@gmail.com'. The 'Advanced parameters' section shows 'Importance' set to 'Importance associated with the email message.' and 'Subject' set to 'Subject of the outgoing email'.

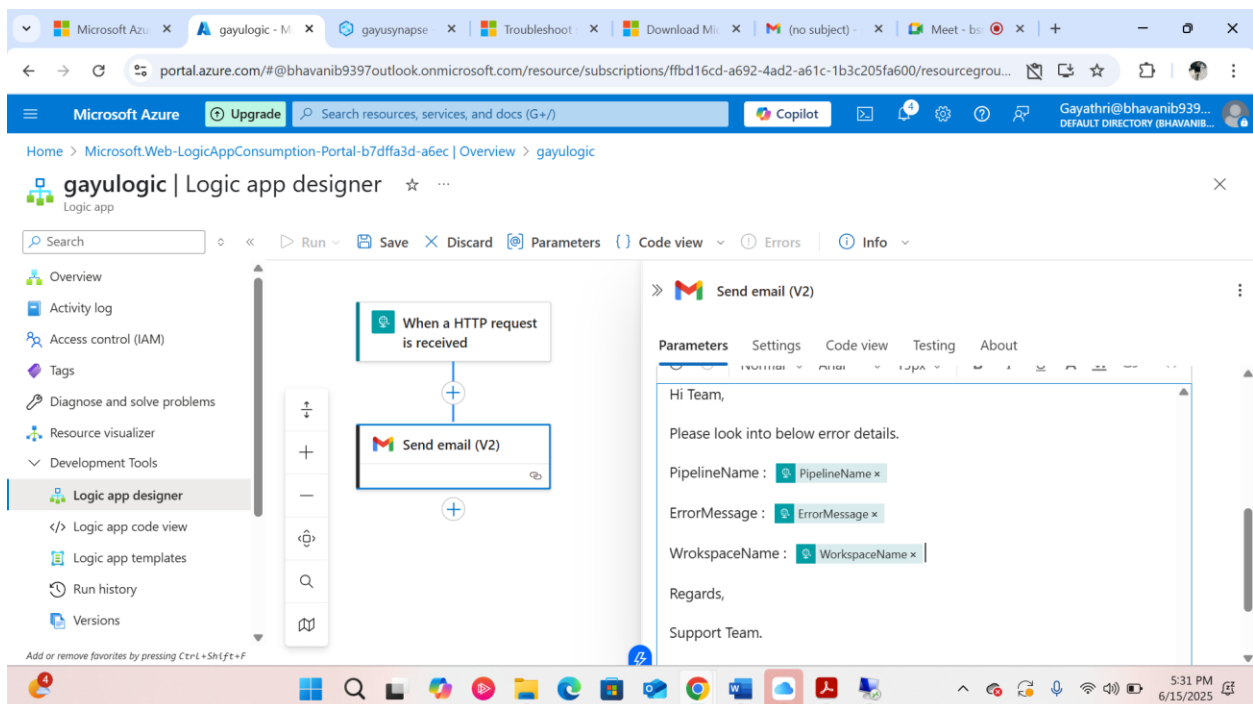
In subject click on first sign near the email and click on pipelineName failedin WorkspaceName

This screenshot shows the same Logic App Designer interface as the previous one, but with the 'Send email (V2)' configuration updated. The 'Subject' field now contains two dynamic expressions: 'PipelineName' followed by 'failedin' and 'WorkspaceName'. The 'Body' field is also visible, containing the text 'Body of the outgoing email.' The 'Importance' field is set to 'High'. The bottom status bar indicates the connection is 'Connected to new_conn_e53c4...'.

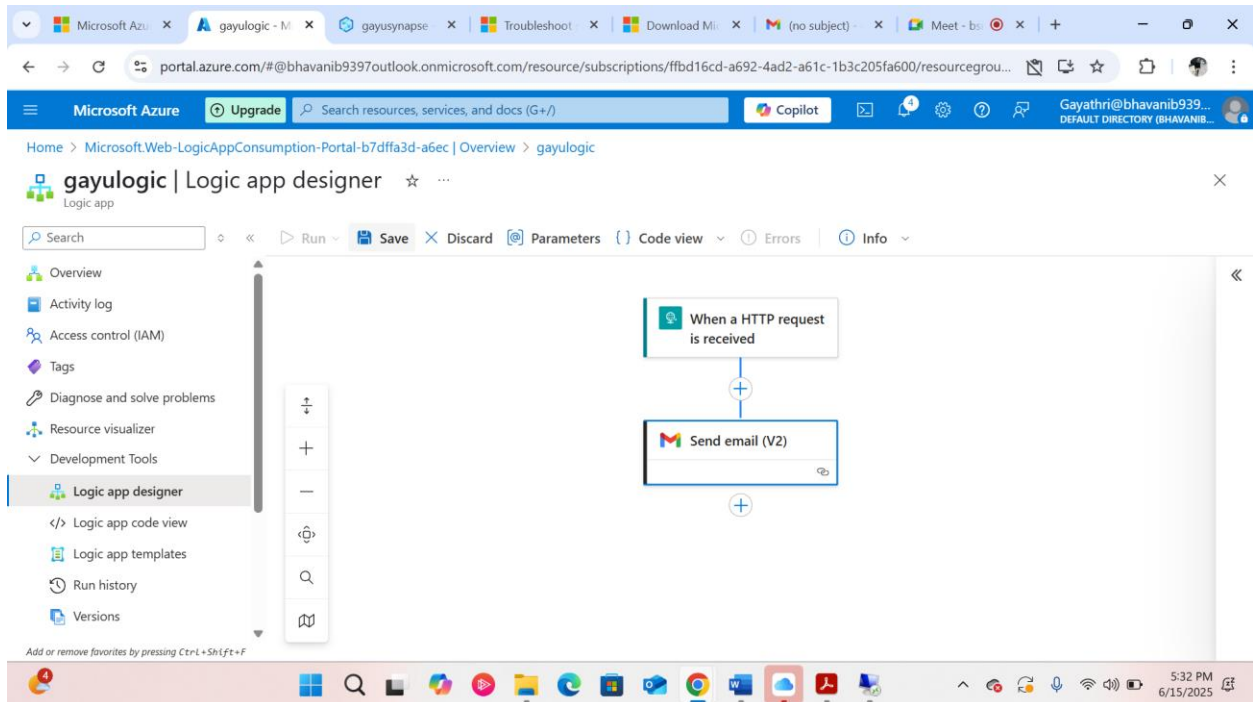
In Body, type the message



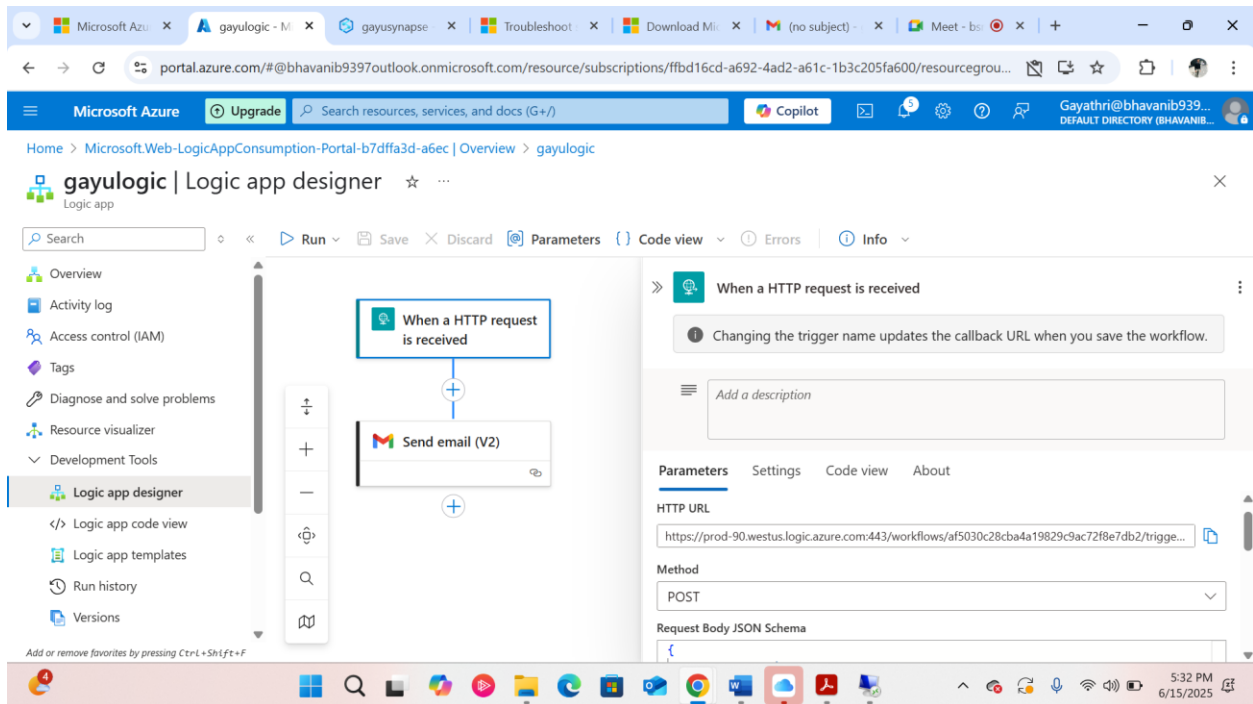
Near the pipeline name, errormessage and workspaceName click the first symbol so it come with symbol



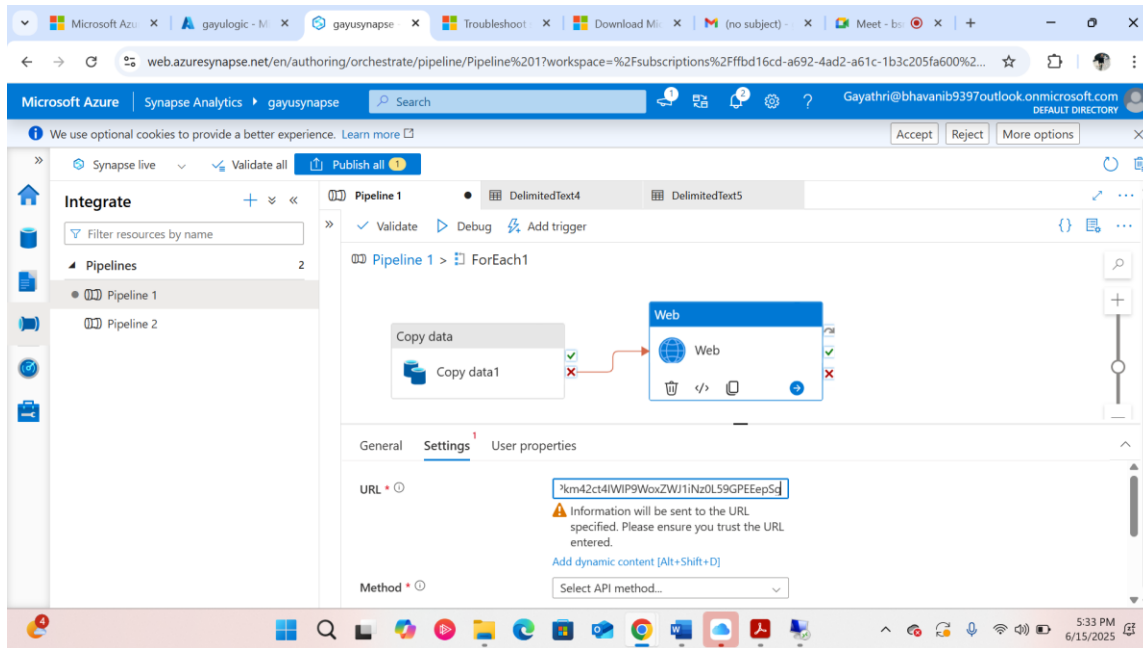
Now, save the logic app



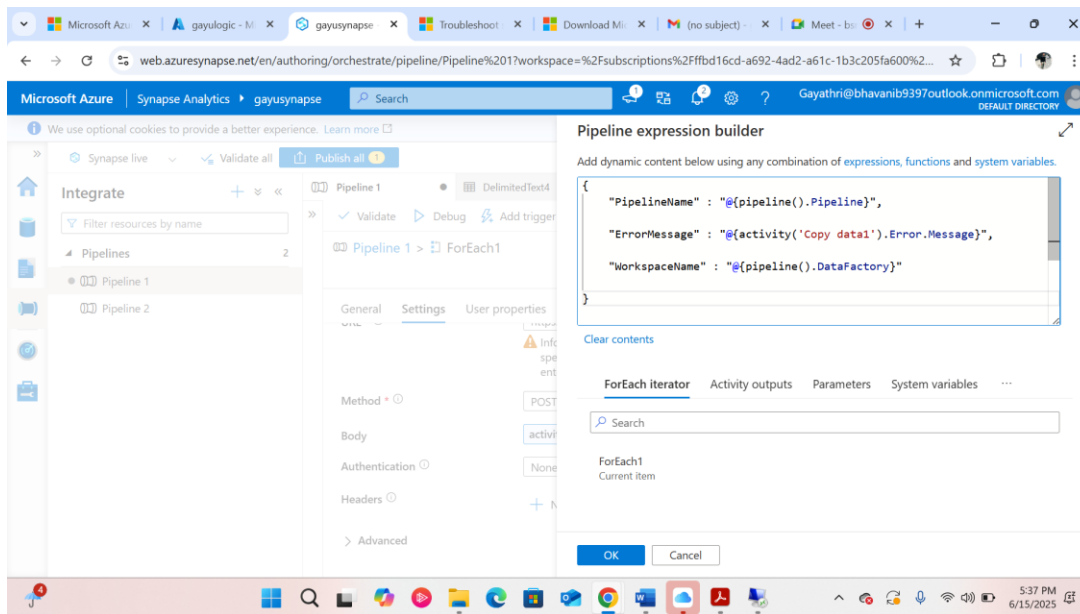
Copy the logic app URL, from parameter



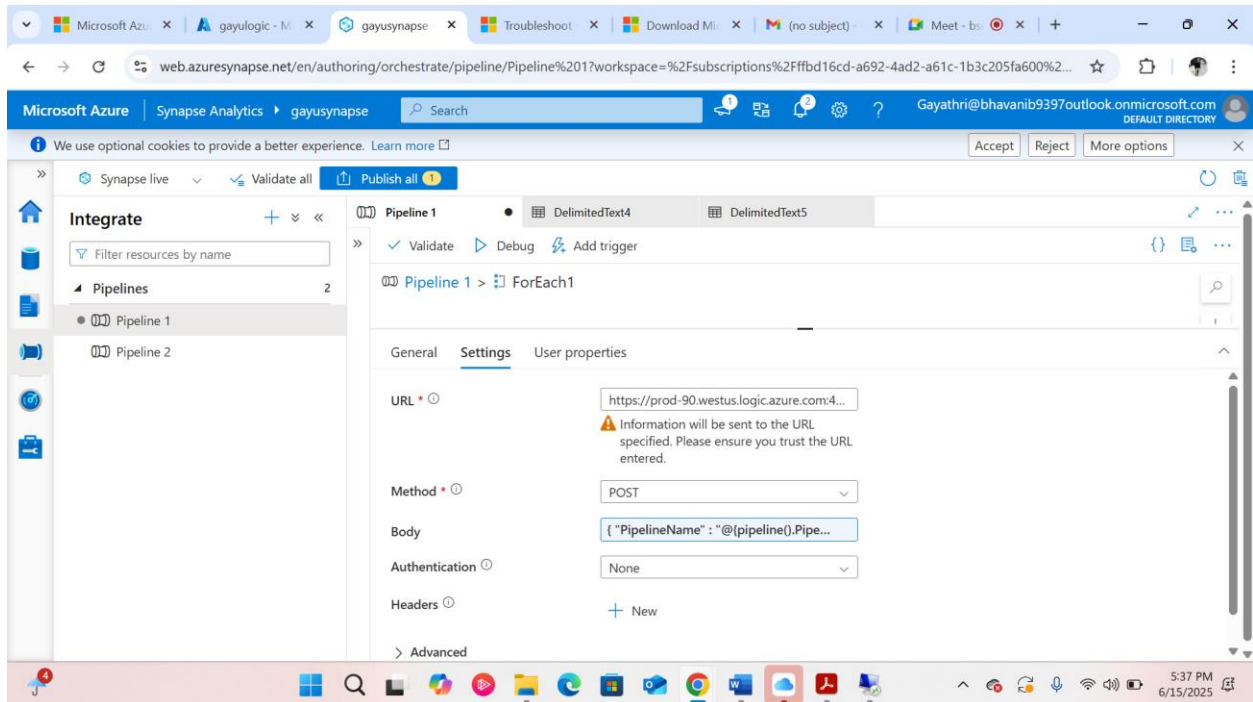
Paste it in Web URL in synapse



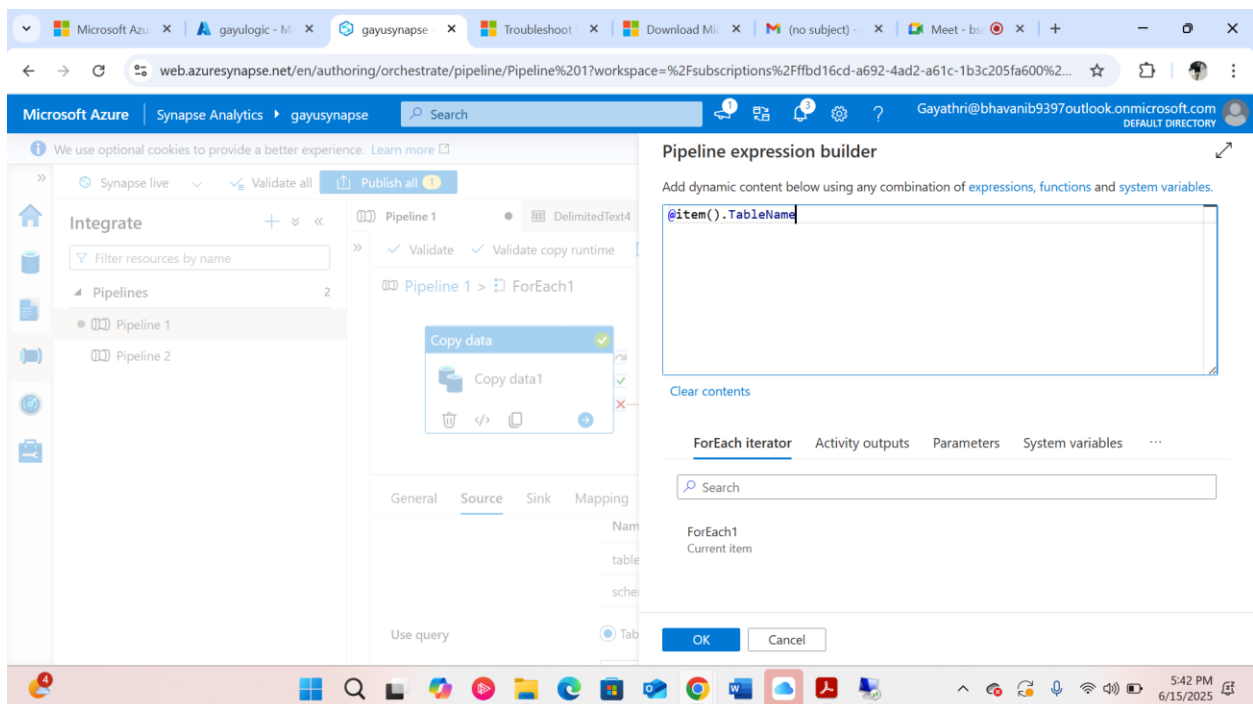
In body, choose the pipeline expression Builder and type the below syntax and click on ok



Method select Post



In pipeline expression for copy data, in source type table name in schema name and click on ok, Copy data will fail.



Copy data1 is failed all 5 table is failed

The screenshot shows the Microsoft Azure Synapse Analytics interface. The left sidebar displays the 'Integrate' section with a list of pipelines. The main area shows the details of 'Pipeline 1', which is currently in a 'Failed' state. The pipeline steps are: 'Lookup' (Succeeded), 'ForEach' (Succeeded), and 'Copy data' (Failed). The 'Copy data' step is failing for all five tables, with the error message 'Copy data1'.

Parameters	Variables	Settings	Output
Web		✓ Succeeded	Web 6/15/2025, 5:42:54 PM 3s AutoResolveIntegrationRi
Web		✓ Succeeded	Web 6/15/2025, 5:42:54 PM 3s AutoResolveIntegrationRi
Copy data1		✗ Failed	Copy data 6/15/2025, 5:42:52 PM Less than 1s Unknown
Copy data1		✗ Failed	Copy data 6/15/2025, 5:42:52 PM Less than 1s Unknown
Copy data1		✗ Failed	Copy data 6/15/2025, 5:42:52 PM Less than 1s Unknown
Copy data1		✗ Failed	Copy data 6/15/2025, 5:42:52 PM Less than 1s Unknown
Copy data1		✗ Failed	Copy data 6/15/2025, 5:42:52 PM Less than 1s Unknown
ForEach1		✓ Succeeded	ForEach 6/15/2025, 5:42:51 PM 8s

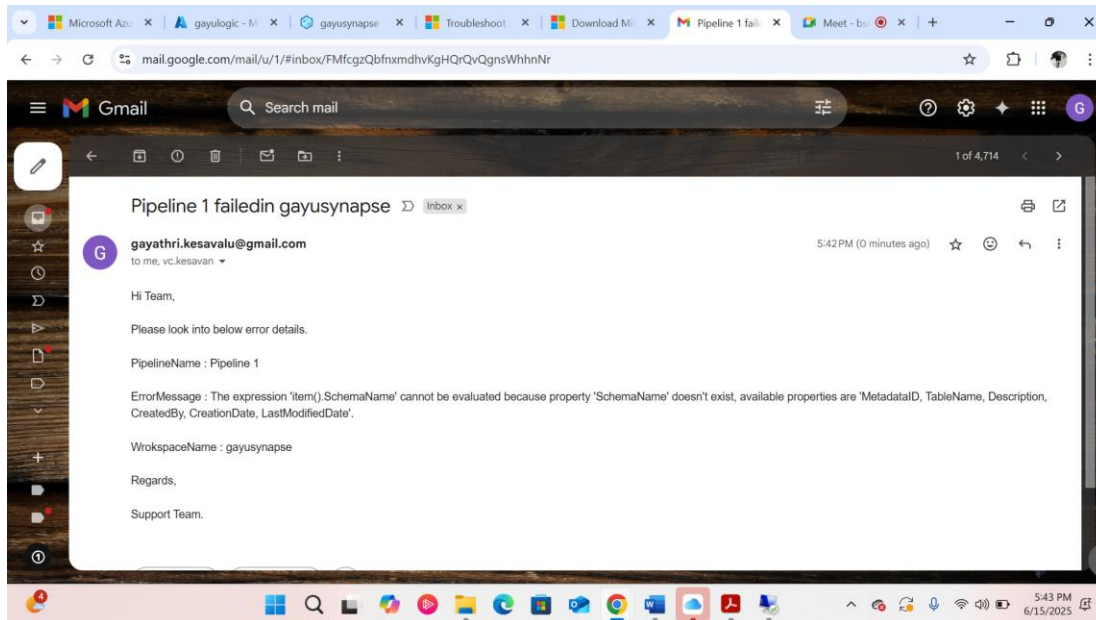
In email, I got a mail that pipeline1 is failed for all 5 table, so it is successful

The screenshot shows a Gmail inbox. The selected email is from 'gayusynapse' with the subject 'Pipeline 1 failedin gayusynapse'. The email body contains the following text:

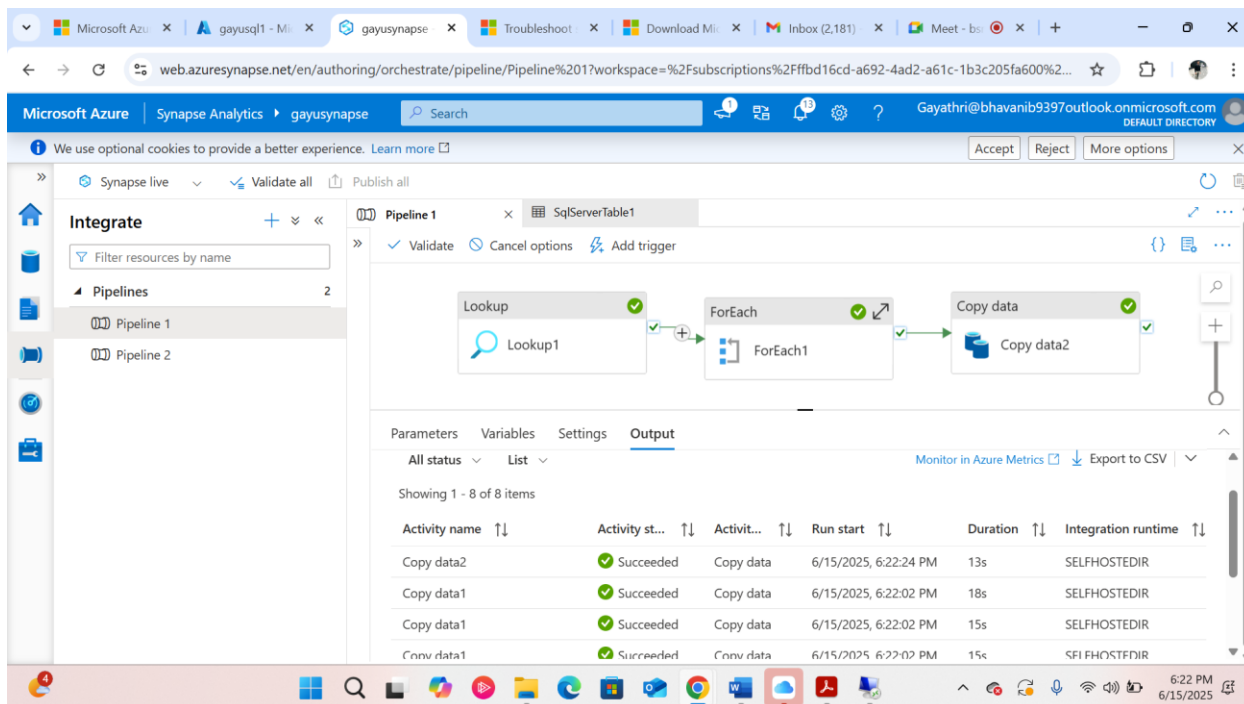
Pipeline 1 failedin gayusynapse - Hi Team, Please look into below error details. PipelineName : Pipeline 1 ErrorMessage ...

The email is dated 5:42 PM. The inbox also shows other emails from 'me', 'Google', 'Uber Eats', and 'The Borrowell Team'.

If, open the failed email, it shows where the error occurred.



Now, copy data is successful, once we fix the error.



Now, I am doing same thing for copy data2 and click on ok after URL and expression is done.

Microsoft Azure | Synapse Analytics | gayusynapse

Search

Gayathri@bhavanib9397outlook.onmicrosoft.com

PIPELINE EXPRESSION BUILDER

Add dynamic content below using any combination of expressions, functions and system variables.

```
{
  "PipelineName" : "@{pipeline().Pipeline}",
  "ErrorMessage" : "@{activity('Copy data2').Error.Message}",
  "WorkspaceName" : "@{pipeline().DataFactory}"
}
```

Clear contents

Activity outputs | Parameters | System variables | Functions | Variables

Search

Copy data1
Copy data1 activity output

Copy data2

OK Cancel

Now, I am running the pipeline, it is successful.

Microsoft Azure | Synapse Analytics | gayusynapse

Search

Gayathri@bhavanib9397outlook.onmicrosoft.com

PIPELINE 1

Lookup1

ForEach1

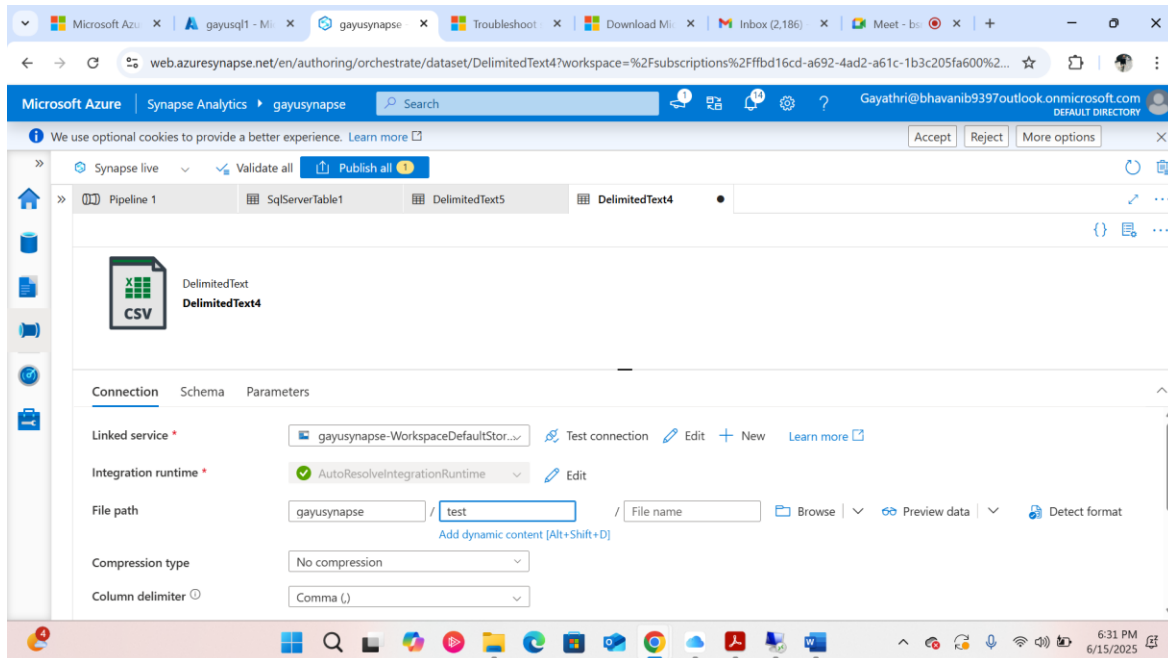
Copy data2

Web2

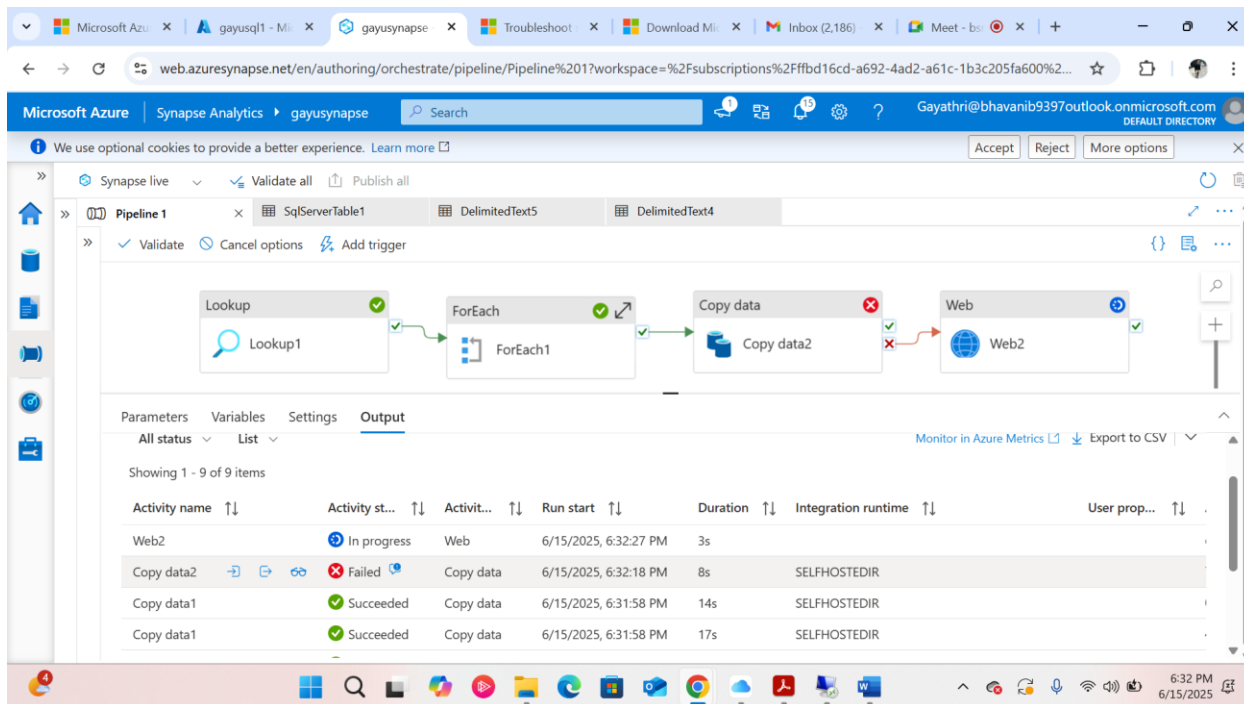
Parameters | Variables | Settings | Output

Activity	Status	Type	Time	Duration	Location
Copy data2	Succeeded	Copy data	6/15/2025, 6:30:05 PM	9s	SELFHOSTEDIR
Copy data1	Succeeded	Copy data	6/15/2025, 6:29:44 PM	15s	SELFHOSTEDIR
Copy data1	Succeeded	Copy data	6/15/2025, 6:29:44 PM	9s	SELFHOSTEDIR
Copy data1	Succeeded	Copy data	6/15/2025, 6:29:44 PM	9s	SELFHOSTEDIR
Copy data1	Succeeded	Copy data	6/15/2025, 6:29:44 PM	15s	SELFHOSTEDIR
Copy data1	Succeeded	Copy data	6/15/2025, 6:29:44 PM	18s	SELFHOSTEDIR
ForEach1	Succeeded	ForEach	6/15/2025, 6:29:43 PM	22s	

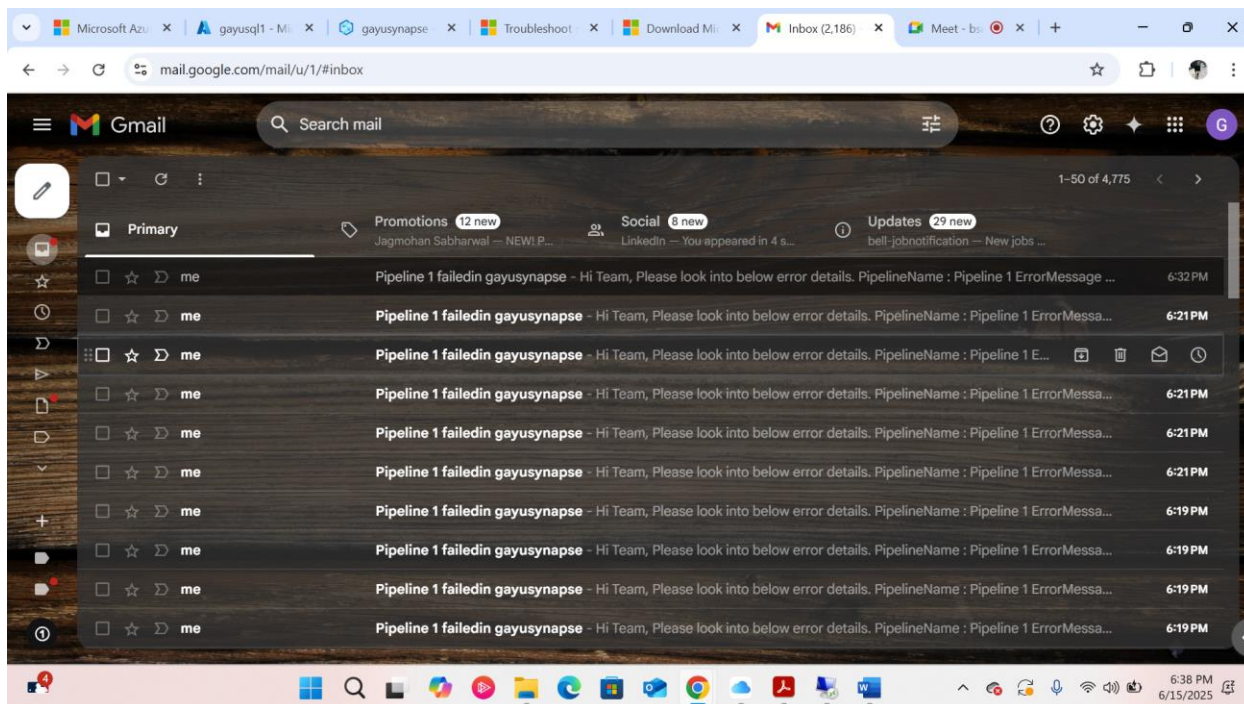
In the folder path I am giving 'test' for throwing error.



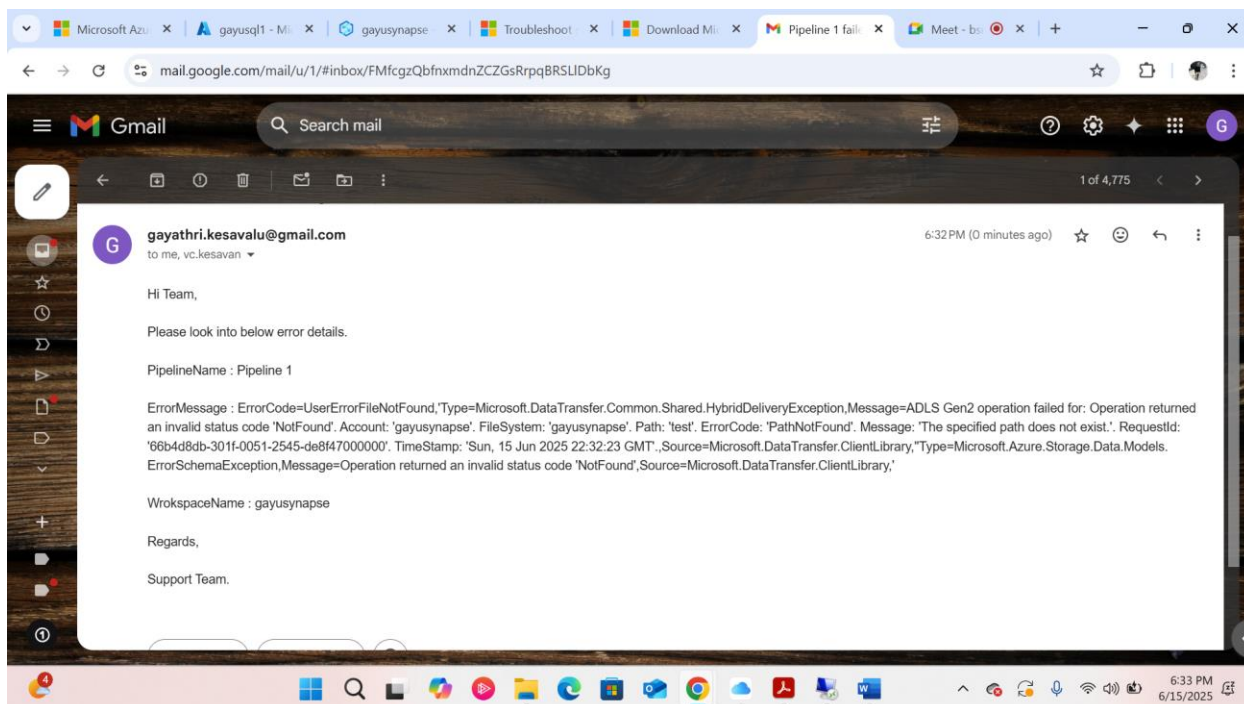
Now, in copydata2 it is failed after publish and debug.



In Email, I can see the failed email, so copy data2 is successful.



I got the email, which place copy data2 failed. So it is successful.



After, fixing the error. Copy data is successful.

The screenshot displays the Microsoft Azure Synapse Analytics web portal. The top navigation bar includes the Microsoft Azure logo, the Synapse Analytics section, and the user's name, Gayathri@bhavanib9397outlook.onmicrosoft.com. A search bar is also present. Below the navigation bar, a cookie consent banner is visible. The main content area shows a pipeline named 'Pipeline 1' with a 'Validate' button and a 'Debug' button. The pipeline is composed of several steps: 'Lookup', 'ForEach', 'Copy data', and 'Web'. The 'Copy data' step is highlighted, and its output is displayed in a table. The table shows that the 'Copy data' step has succeeded multiple times, with the output being 'Copy data' and the destination being 'SELFHOSTEDIR'. The table also shows the execution time for each step, with the 'Copy data' step taking 15s, 9s, 15s, and 9s respectively. The 'Lookup' step took 11s, and the 'ForEach' step took 19s. The table is organized into columns: Parameters, Variables, Settings, and Output. The 'Output' column shows the status of each step, with all steps marked as 'Succeeded'.

Parameters	Variables	Settings	Output
Copy data1		✓ Succeeded	Copy data 6/15/2025, 6:33:46 PM 15s SELFHOSTEDIR
Copy data1		✓ Succeeded	Copy data 6/15/2025, 6:33:47 PM 9s SELFHOSTEDIR
Copy data1		✓ Succeeded	Copy data 6/15/2025, 6:33:46 PM 9s SELFHOSTEDIR
Copy data1		✓ Succeeded	Copy data 6/15/2025, 6:33:47 PM 15s SELFHOSTEDIR
Copy data1		✓ Succeeded	Copy data 6/15/2025, 6:33:46 PM 9s SELFHOSTEDIR
Copy data1		✓ Succeeded	Copy data 6/15/2025, 6:33:46 PM 9s SELFHOSTEDIR
ForEach1		✓ Succeeded	ForEach 6/15/2025, 6:33:46 PM 19s
Lookup1		✓ Succeeded	Lookup 6/15/2025, 6:33:34 PM 11s SELFHOSTEDIR